DESIGN COMPENDIUM FOR FLOOD RESILIENT HOUSING

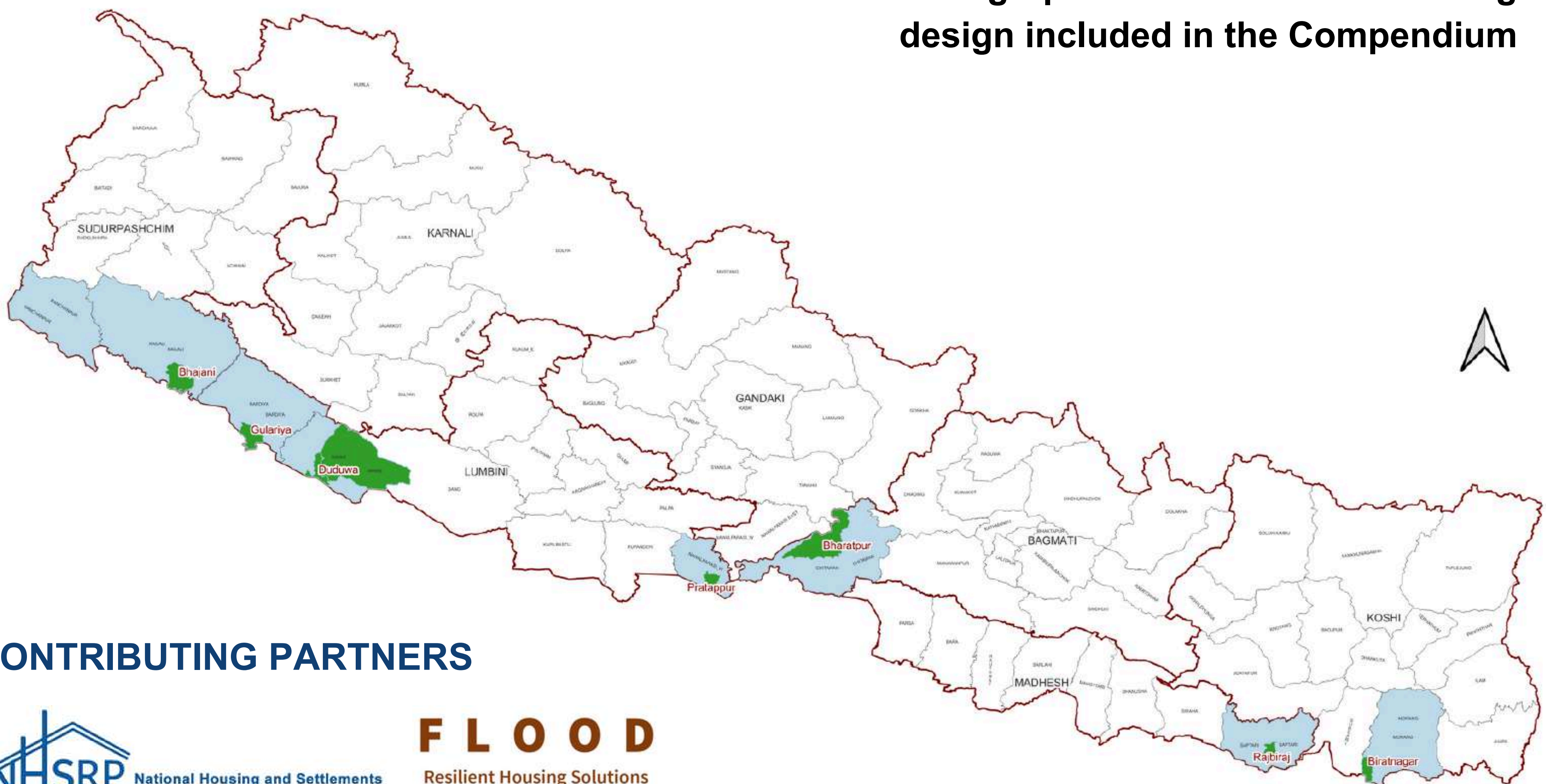
HOMES AND COMMUNITIES

1st Edition

NHSRP National Housing and Settlements Resilience Platform

FLOOD Resilient Housing Solutions Technical Working Group

Geographic distribution of housing design included in the Compendium



CONTRIBUTING PARTNERS

FOREWORD

Floods have become a recurring hazard in Nepal, particularly affecting the Terai region where communities are regularly displaced and homes are destroyed. Addressing these challenges requires innovative, practical solutions, and this **Flood Resilient Housing Compendium** is a step towards meeting that need.

This publication is the first of its kind, showcasing housing designs that have incorporated essential flood-resilient features. It is intended to support a wide range of users, from local governments to non-governmental organizations, architects, and community leaders, in adopting designs that offer protection and sustainability.

We commend the efforts of the Flood Technical Working Group and its partners in creating this valuable resource. It is our sincere hope that these designs will inspire further developments and foster long-term resilience in flood-prone communities across Nepal.

PREFACE

The Flood Resilient Housing Design Compendium is developed in response to the increasing frequency and intensity of floods in Nepal, particularly in the Terai region, which has experienced significant damage to infrastructure, homes, and livelihoods. This Compendium provides a collection of nine housing designs, each incorporating flood-resilient measures that aim to mitigate the risks posed by future flooding events. The Compendium was developed by efforts of Flood Resilient Housing Solutions: Technical Working Group's "Sub-group on Development of Design Compendium".

This compilation has been made possible through the collaboration of various partner organizations and stakeholders who have contributed their expertise and designs. It is our hope that this Compendium will serve as a foundational resource for architects, engineers, policymakers, and local communities seeking to build more resilient housing solutions in flood-prone areas.

We would like to express our gratitude to the Flood Technical Working Group and our partner organizations for their invaluable contributions to this project.

COMPENDIUM DEVELOPMENT COMMITEE

Sub-group: Development of Flood-Resilient Housing Design Compendium

- Rupesh Shrestha (NHSRP, Lead Flood : Technical Working Group)
- Aashish Gautam (Build-up Nepal)
- Sujata Kayastha (Tukee Foundation)
- Rabin Chaulagain (NHSRP)
- Shramina Shrestha (NHSRP Consultant)
- Jyoti Bhandari (Habitat for Humanity Nepal)
- Bikash Karna (Caritas Nepal)

Review Team

- From NHSRP: Rajesh Sunuwar, Bibek Shrestha, Ambika Amatya, Reshma Shrestha, Birendra Shrestha, Prakash Basnet
- From POs: Kajal Pradhan, Indra Dev Chaurasiya, Narendra K.C., Janak Sharma, Sheela Sen Thakuri, Punita Chaudhary

CONTRIBUTING PARTNERS

NHSRP National Housing and Settlements Resilience Platform

FLOOD Resilient Housing Solutions Technical Working Group

CRS

Caritas Nepal

BUILD UP NEPAL

Habitat for Humanity Nepal

Tukee Foundation

CHITWAN ARCHITECTURE DESIGN STUDIO

TABLE OF CONTENTS

INTRODUCTION

Floods are a recurrent natural hazard in Nepal, particularly in the Terai region and other low-lying areas. The annual monsoon rains often lead to rivers overflowing, causing significant damage to homes, livelihoods, and infrastructure. The vulnerability of Nepalese housing, especially in rural areas, makes flood resilience an urgent priority.

This catalog presents nine flood-resilient housing designs, each incorporating structural and non-structural elements that enhance the ability of homes to withstand flooding. These designs are not intended to be prescriptive guidelines but rather illustrative examples that demonstrate potential strategies for building safer, more resilient homes in flood-prone areas.

Each design has been assessed for its suitability in the Nepalese context, considering factors such as local materials, construction techniques, and cultural preferences. Additionally, these designs integrate modern flood-resilience strategies, such as raised plinths, stilts, and floodproof materials, offering practical solutions for diverse geographic and socioeconomic conditions. All designs have been constructed at locations specified.

SCOPE

The “Flood Resilient Housing Compendium” has been designed for a wide range of users, including architects, engineers, local governments, community organizations, and builders involved in the planning and construction of flood-resilient homes in Nepal.

This Compendium includes nine housing designs, each developed with specific attention to the flood-prone regions of the Terai. The designs address key flood resilience factors, such as elevated foundation types, modular construction methods, and material choices that enhance the durability and safety of homes in flood-affected areas.

While the primary focus is on housing in the Terai, the principles outlined in this Compendium can be adapted to other flood-prone regions of Nepal. The Compendium offers a starting point for integrating resilient housing solutions into disaster risk reduction strategies.

LIMITATIONS

1. Non-Exhaustive Designs: The compendium presents only nine housing designs, which may not encompass the full range of flood resilience strategies or cover all flood-prone regions of Nepal.
2. Adaptability to Local Conditions: While the designs are tailored to the Nepalese context, site-specific considerations such as soil conditions, flood history, and local material availability may require further adaptation.
3. Guidance, Not Regulations: The compendium is intended to inspire and inform, but it does not constitute official building codes or standards. Builders and designers must consider local regulations and seek professional consultation where necessary.
4. Cost and Accessibility: The feasibility of constructing flood-resilient homes may vary based on the availability of materials, labor, and financial resources, which are not comprehensively addressed in this catalog.
5. Ongoing Research: Flood-resilient housing design is a field of continuous evolution, and future developments in materials, technologies, and flood modeling may offer new insights not covered in this document.

FLOOD RESILIENT HOUSE TYPE DESIGN-I:

TIMBER FRAME WITH LATH AND PLASTER WALL CONSTRUCTION

INTRODUCTION

House Type: Timber Frame with Lath and Plaster Wall Construction

Designed By: Catholic Relief Services (CRS) and Caritas Nepal

Project Title: Flood Resilient Demo-house

Project Location:

1. Province: Lumbini
2. District: Banke
3. Ward: 01
4. Municipality: Rapti Sonari Rural Municipality

LOCATION MAP

CLIMATIC FEATURES:

Avg. Maximum Annual Temperature: 32-35˚C

Avg. Minimum Annual temperature: 8-10˚C

Annual Rainfall: 1500-2000mm, mostly in June to September

Average relative Humidity: 60-80%

GEOTECHNICAL FEATURES:

Topography: Flat with lowland riverbanks

Mean Sea Level (MSL): 150-300 meters approx.

Soil Characteristics: Primary Alluvial soil

Disaster: Seasonal flooding

WHY IS THIS DESIGN FLOOD RESILIENT?

Features of this design includes:

1) Raised plinth level

2) Lime Stabilized Soil Technology

3) Better connection detailing of wood, metal and concrete

4) Earthquake Resilient Technology

5) Use of local natural building materials

Lime Stabilized Soil (LSS) is a soil made up of quick lime (CaO) and other admixtures.

The use of this technology can increase the load-bearing capacity of soil, improve the stability of the soil under water, and does not allow the soil to be washed away during floods. Local skills and maximum use of local materials are added benefits of this technology. The building has been inundated with flood water in Sept 28th 2024 and July 6th 2024. Structural elements; plasters and flooring are intact.

TIMBER FRAME WITH LATH AND PLASTER WALL CONSTRUCTION

DESIGN CONSIDERATIONS

Available Building Materials: Stones, timber, cement, sand, aggregate. quick lime, straw, rice husk ash

Foundation: Stone Masonry with cement mortar

Plinth: concrete band

Post: Timber

Fence/Wall: Lath and Plaster

Openings: Plywood, Timber

Joints: Nuts and Bolt

Treatment (wood): Seasoning of Salwood

Roof Type: Two way slope

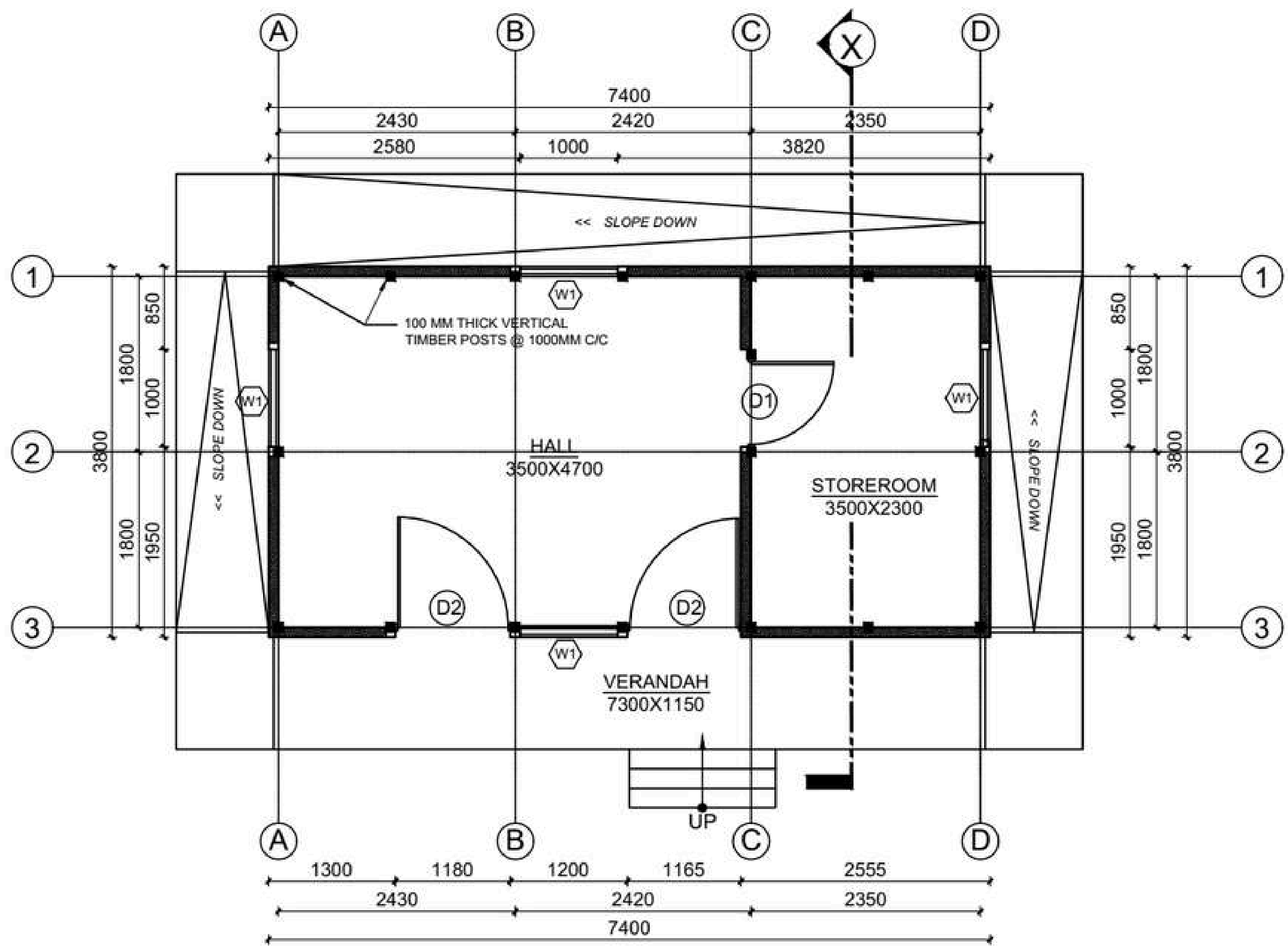
Roof cover: CGI Sheet

Roof structure: Timber

Bracing: Timber

Note: Rate analysis based on field experience

LEVEL	MANPOWER		MATERIAL							
	SKILLED	UNSKILLED	STONE	MUD	CEMENT	SAND	AGGREGATE	WOOD	CGI SHEET	REFORCEMENT BARS
	Md	Md	Cu.m	Cu.m	Bags	Cu.m	Cu.m	Cu.m	Bundle	Kg
UP TO PLINTH LVL	49	177	30	25	95	12	3.5	-	-	380
SUPERSTRUCTURE	65	119	-	49	-	0.5	-	86.9	-	-
ROOFING	65	-	-	-	-	-	-	-	2.86	-
TOTAL	**114**	**296**	**30**	**74**	**95**	**12.5**	**3.5**	**86.9**	**2.86**	**380**



A **GROUND FLOOR PLAN**
AREA: 22.57 SQ.M

DESIGNED BY:

PROJECT TITLE:
FLOOD RESILIENT DEMO-HOUSE

DESIGN TYPE:
TIMBER STRUCTURE WITH LATH & PLASTER WALL CONSTRUCTION

DRAWING TITLE:
FLOOR PLAN

LOCATION:
RAPTI SONARI, NEPAL

DRAWN BY:
CRS, CARITAS NEPAL

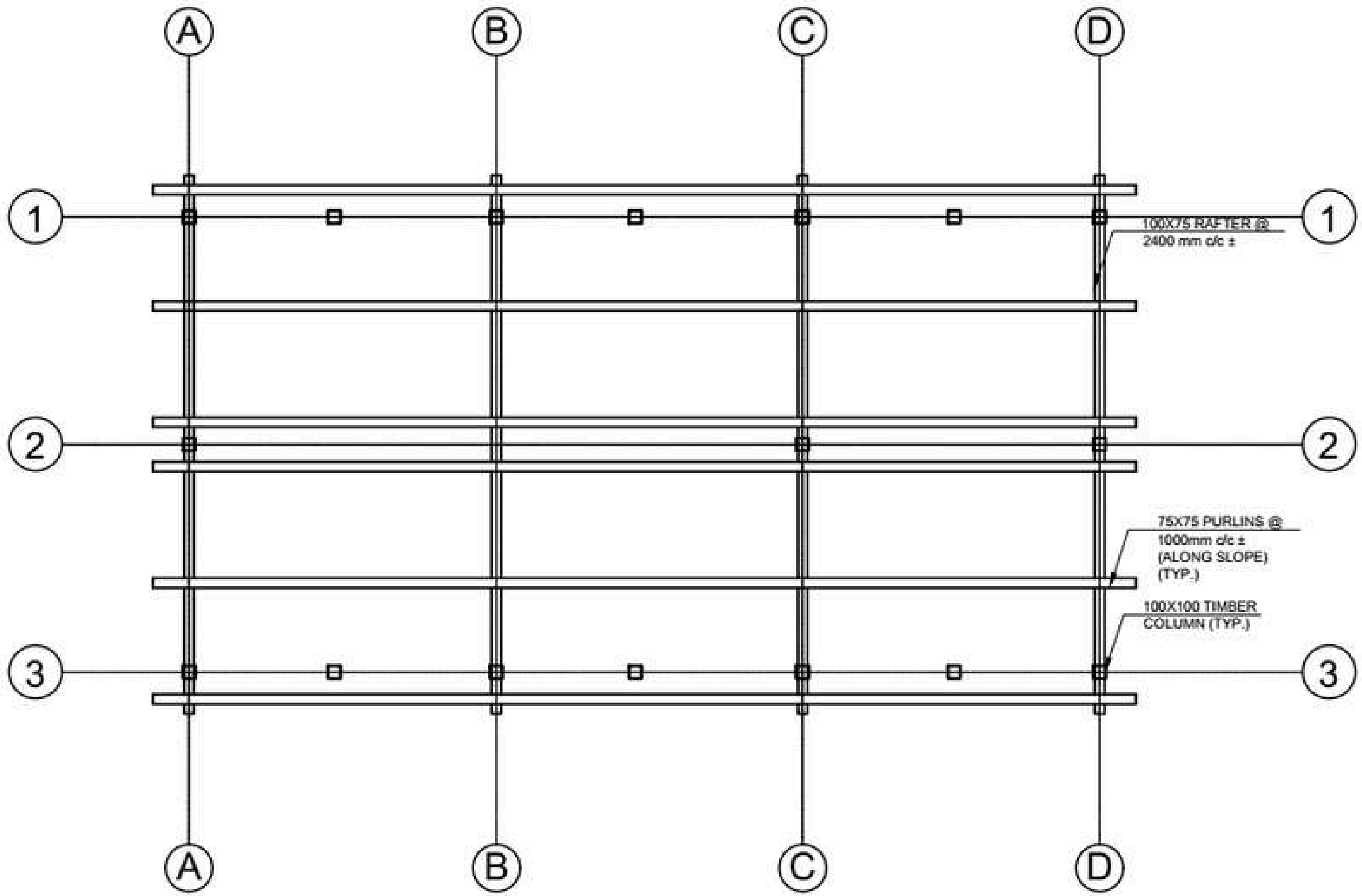
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	900 X 2200	3
2.	WINDOW	W1	900 X 1200	4

SCALE: NOT TO SCALE

DATE: **2023 AD**

SHEET NO. **A-01**



A **ROOF FRAME PLAN**
SCALE: 1:50

DESIGNED BY:

PROJECT TITLE:
FLOOD RESILIENT DEMO-HOUSE

DESIGN TYPE:
TIMBER STRUCTURE WITH LATH & PLASTER WALL CONSTRUCTION

DRAWING TITLE:
FLOOR PLANS

LOCATION:
RAPTI SONARI, NEPAL

DRAWN BY:
CRS, CARITAS NEPAL

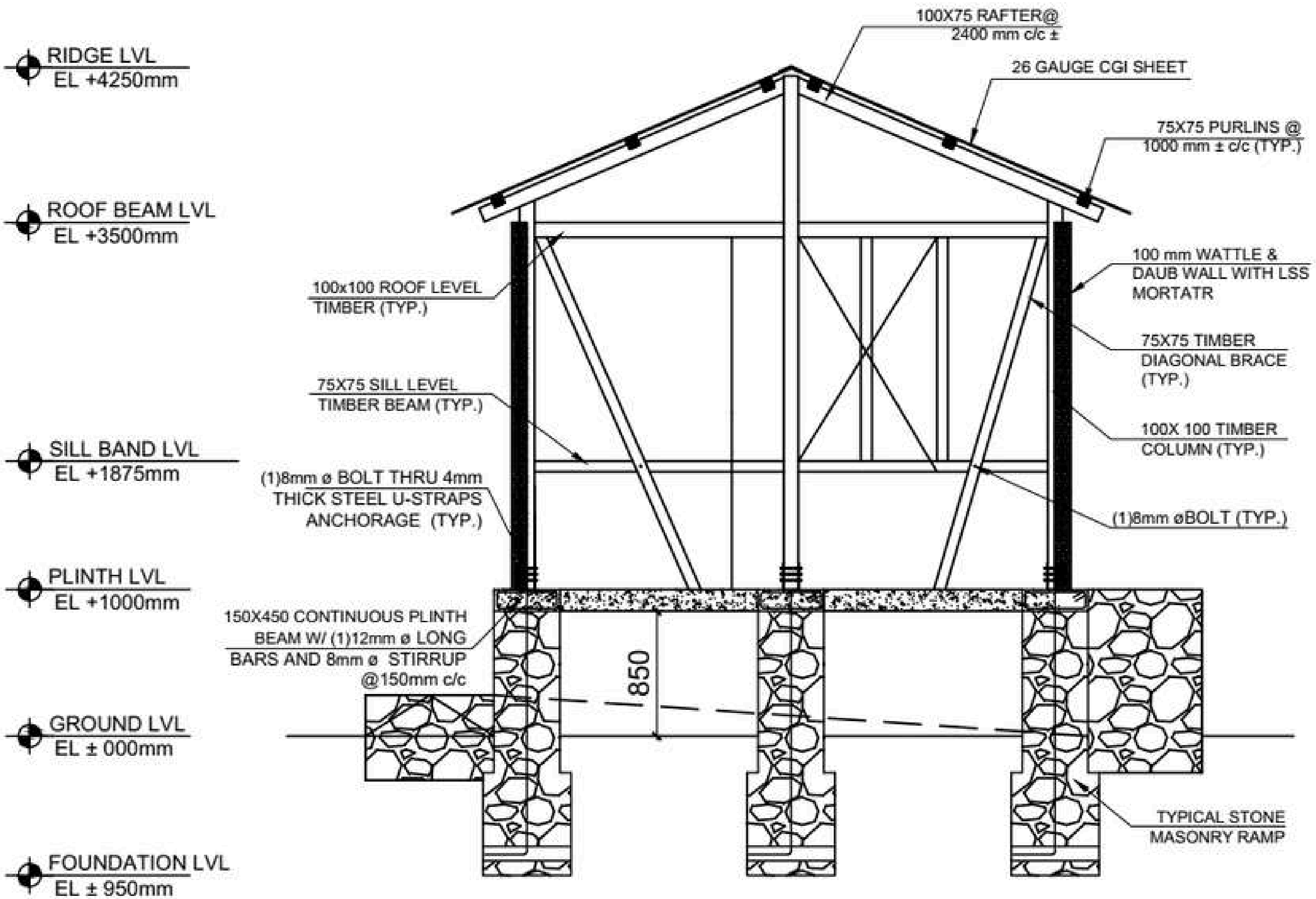
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	900 X 2200	3
2.	WINDOW	W1	900 X 1200	4

SCALE:
NOT TO SCALE

DATE:
2023 AD

SHEET NO.
A-02



A **SECTION AT X-X**
SCALE: 1:40

DESIGNED BY:

PROJECT TITLE:
FLOOD RESILIENT DEMO-HOUSE

DESIGN TYPE:
TIMBER STRUCTURE WITH LATH & PLASTER WALL CONSTRUCTION

DRAWING TITLE:
SECTION AT X-X

LOCATION:
RAPTI SONARI, NEPAL

DRAWN BY:
CRS, CARITAS NEPAL

N E S W

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	900 X 2200	3
2.	WINDOW	W1	900 X 1200	4

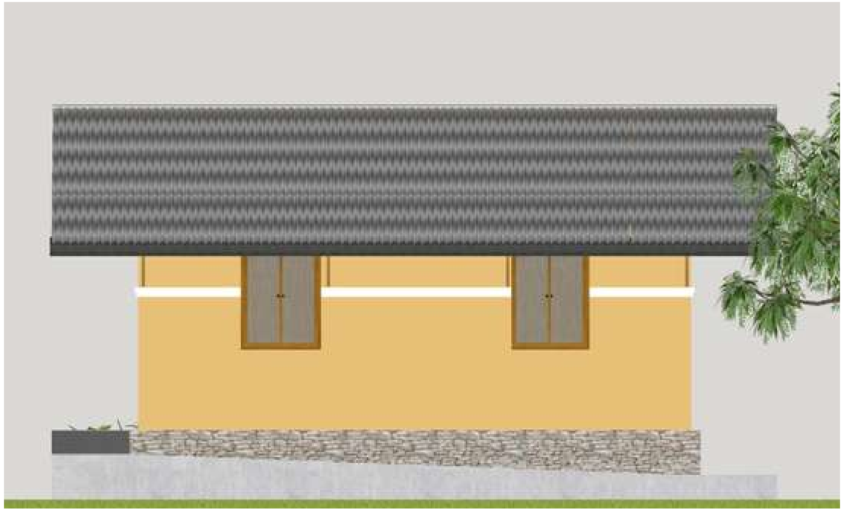
SCALE:
NOT TO SCALE

DATE:
2023 AD

SHEET NO.
A-03

FRONT ELEVATION

SIDE ELEVATION



BACK ELEVATION

SIDE ELEVATION

DESIGNED BY:

CRS

PROJECT TITLE:
FLOOD RESILIENT DEMO-HOUSE

DESIGN TYPE:
TIMBER STRUCTURE WITH LATH & PLASTER WALL CONSTRUCTION

DRAWING TITLE:
ELEVATION

LOCATION:
RAPTI SONARI, NEPAL

DRAWN BY:
CRS, CARITAS NEPAL

N
W E
S

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	900 X 2200	3
2.	WINDOW	W1	900 X 1200	4

SCALE:
NOT TO SCALE

DATE:
2023 AD

SHEET NO.
A-03

FLOOD RESILIENT HOUSE TYPE DESIGN II:

BAMBOO FRAME WITH WATTLE AND DAUB WALL CONSTRUCTION

INTRODUCTION

House Type: Bamboo Frame with Wattle and Daub Lime Stabilized Soil Infill

Designed By: Catholic Relief Services (CRS) and Caritas Nepal

Project Title: Flood Resilient Demo-house

Project Location:

1. Province: Lumbini
2. District: Banke
3. Ward: 1
4. Municipality: Duduwa Rural Municipality

LOCATION MAP

CLIMATIC FEATURES:

Avg. Maximum Annual Temperature: 30-35˚C

Avg. Minimum Annual temperature: 15-20˚C

Annual Rainfall: 1400mm

Average relative Humidity: 60-80%

GEOTECHNICAL FEATURES:

Topography: Flat with lowland riverbanks

Mean Sea Level (MSL): 200-300 meters approx.

Soil Characteristics: Fertile Alluvial soil

Disaster: Seasonal flooding

WHY IS THIS DESIGN FLOOD RESILIENT?

Features of this design includes:

1) Raised plinth level

2) Lime Stabilized Soil Technology

3) Better connection detailing of wood, metal and concrete

4) Earthquake Resilient Technology

5) Use of local natural building materials

Lime Stabilized Soil (LSS) is a soil made up of quick lime (CaO) and other admixtures.

The use of this technology can increase the load-bearing capacity of soil, improve the stability of the soil under water, and does not allow the soil to be washed away during floods. Local skills and maximum use of local materials are added benefits of this technology. In September 2024, flooding occurred in Duduwa Rural Municipality well below the elevated platform.

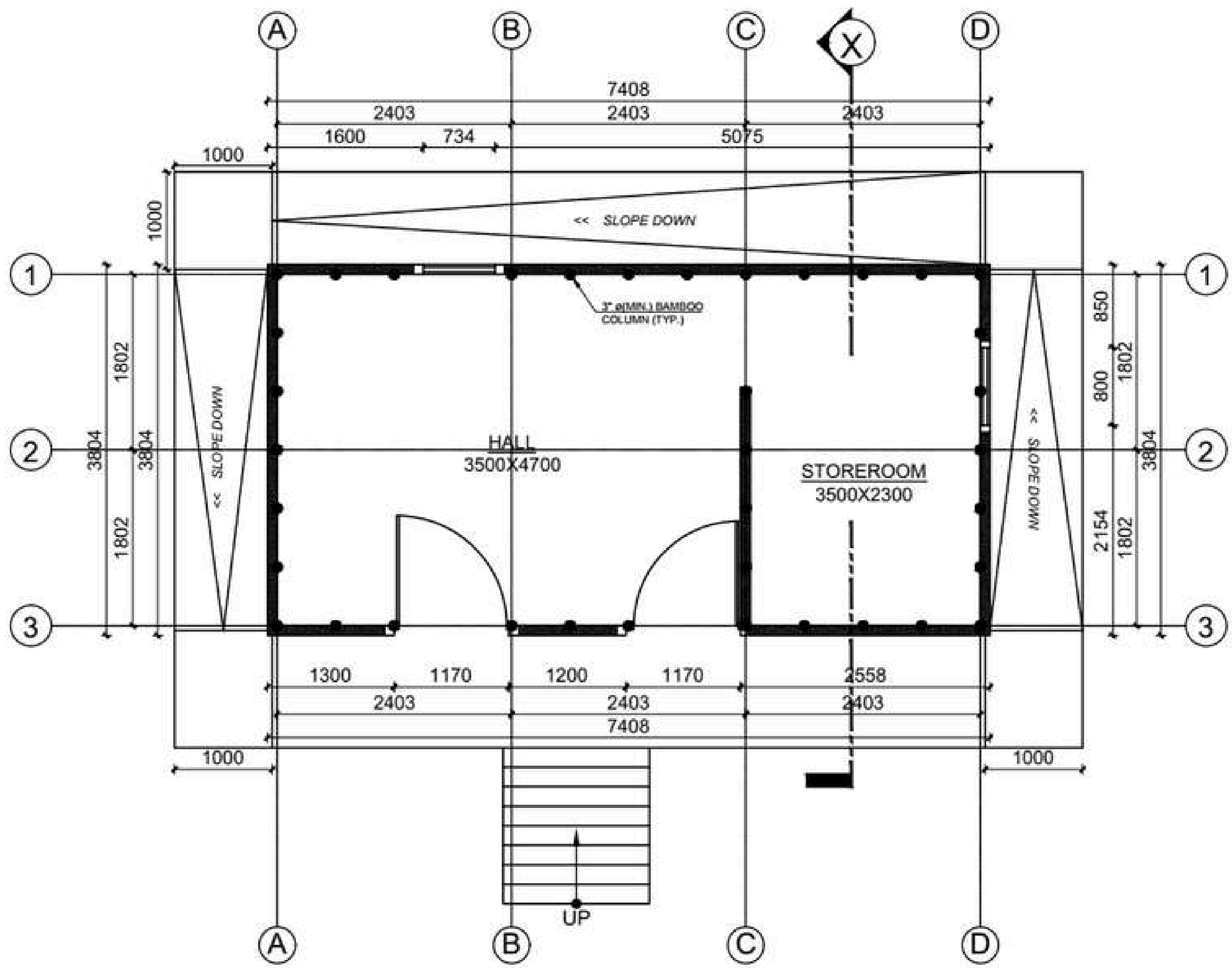
BAMBOO FRAME WITH WATTLE AND DAUB WALL CONSTRUCTION

DESIGN CONSIDERATIONS

Available Building Materials: brick, bamboo, cement, sand, aggregate. quick lime, straw, rice husk
Foundation: Brick Masonry with cement mortar
Plinth: PPC
Post: Bamboo
Fence/Wall: Wattle and Daub
Openings: Plywood, Bamboo
Joints: Knot Bolt
Treatment (wood): Chemical treatment of Bamboo with Borax/ Boric Acid
Roof Type: Two way slope
Roof cover: CGI Sheet
Roof structure: Bamboo
Bracing: Bamboo

Note: Rate analysis based on field experience

LEVEL	MANPOWER		MATERIAL								
	SKILLED	UNSKILLED	STONE	BRICK	CEMENT	SAND	AGGREGATE	BAMBOO	WOOD	CGI SHEET	REFORCEMENT BARS
	Md	Md	Cu.m	No.s	Bags	Cu.m	Cu.m	No.s	Cu.m	Bundle	Kg
UP TO PLINTH LVL	4	36	39.67	1528	11	26.3	1.5	-	-	-	-
SUPERSTRUCTURE	59	127	-	12442	44	6.59	-	50	0.18	-	387.43
ROOFING	18	24	-	-	-	-	-	52	-	4	-
TOTAL	**81**	**187**	**39.67**	**13970**	**55**	**32.89**	**1.5**	**102**	**0.18**	**4**	**387.43**



GROUND FLOOR PLAN
AREA: 33.21 SQ.M

DESIGNED BY:

PROJECT TITLE:
Flood Resilient Demo-house

DESIGN TYPE:
BAMBOO STRUCTURE WITH WATTLE & DAUB WALL CONSTRUCTION

DRAWING TITLE:
FLOOR PLANS

LOCATION:
Duduwa, NEPAL

DRAWN BY:
CRS, CARITAS NEPAL

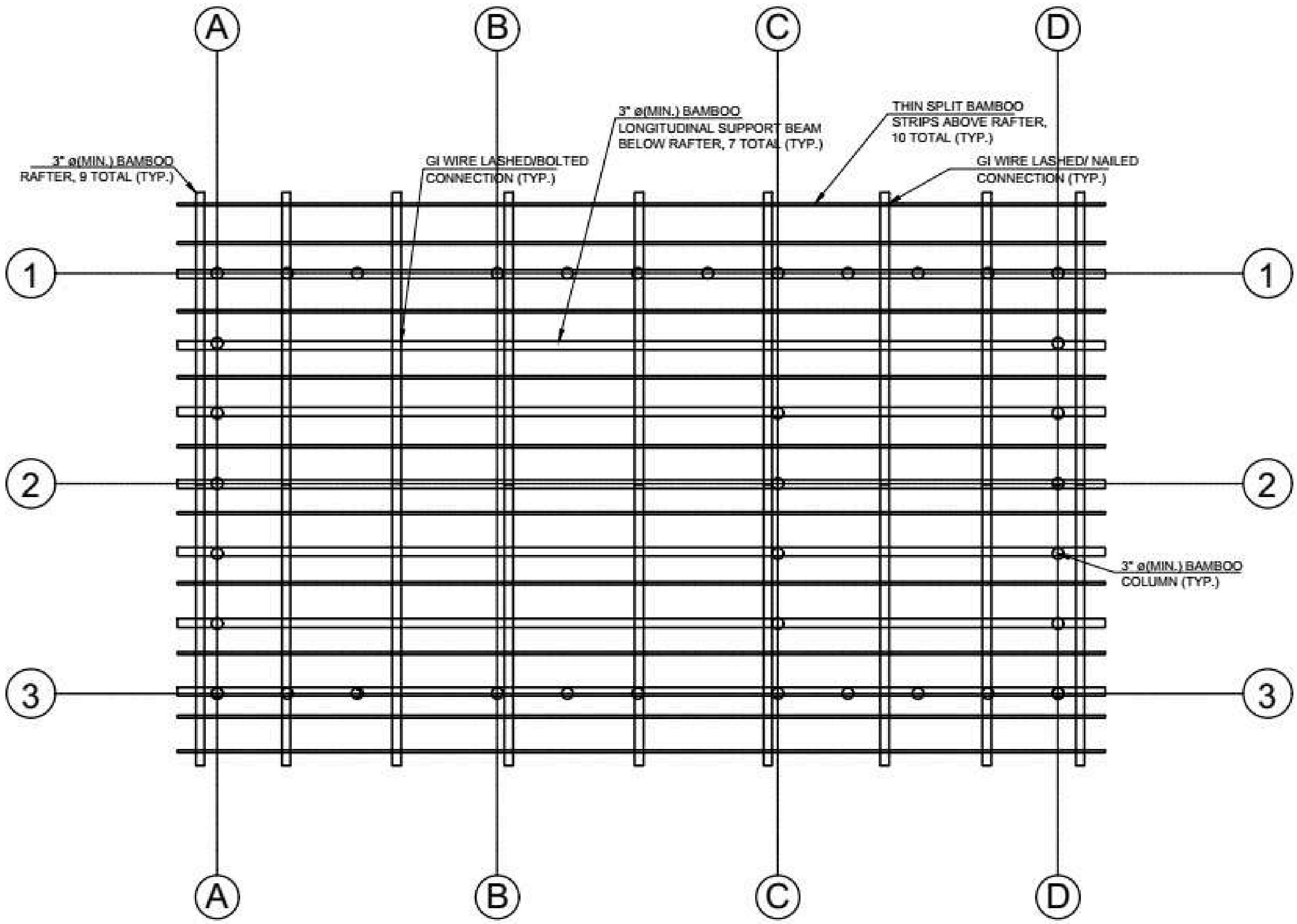
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	900 X 2200	3
2.	WINDOW	W1	900 X 1200	4

SCALE: NOT TO SCALE

SHEET NO. **A-01**

DATE: **2079/80 AD**



A ROOF FRAME PLAN
SCALE: 1:50

DESIGNED BY:
CRS

PROJECT TITLE:
Flood Resilient Demo-house

DESIGN TYPE:
BAMBOO STRUCTURE WITH WATTLE & DAUB WALL CONSTRUCTION

DRAWING TITLE:
FLOOR PLANS

LOCATION:
DUDUWA, NEPAL

DRAWN BY:
CRS, CARITAS NEPAL

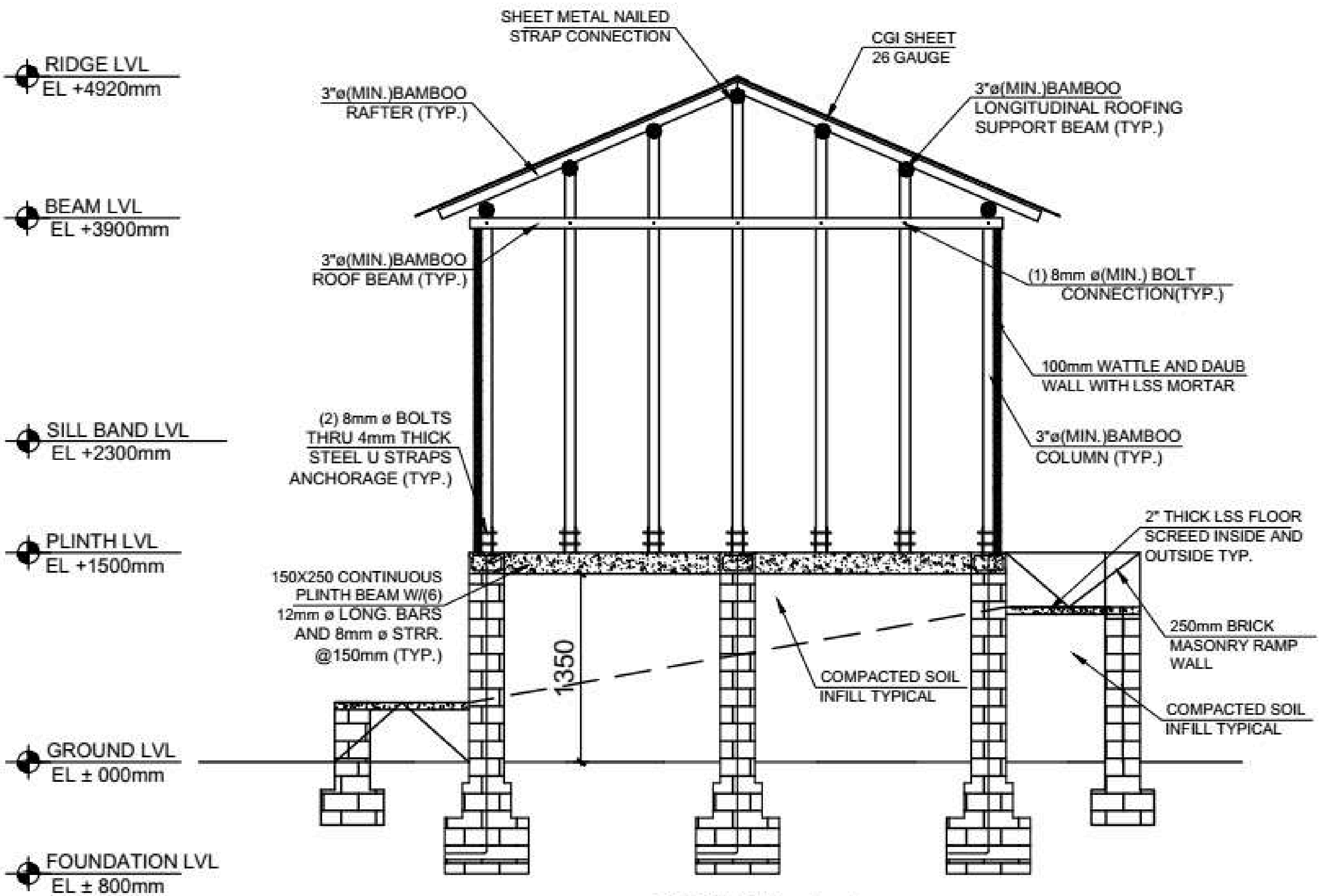
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	900 X 2200	3
2.	WINDOW	W1	900 X 1200	4

SCALE:
NOT TO SCALE

DATE:
2079/80 AD

SHEET NO.
A-02



SECTION 1—1

A SECTION AT X-X
SCALE: 1:40

DESIGNED BY:
CRS

PROJECT TITLE:
Flood Resilient Demo-house

DESIGN TYPE:
TIMBER STRUCTURE WITH LATH & PLASTER WALL CONSTRUCTION

DRAWING TITLE:
FLOOR PLANS

LOCATION:
Duduwa, Nepal

DRAWN BY:
CRS, CARITAS NEPAL

N W E S

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	900 X 2200	3
2.	WINDOW	W1	900 X 1200	4

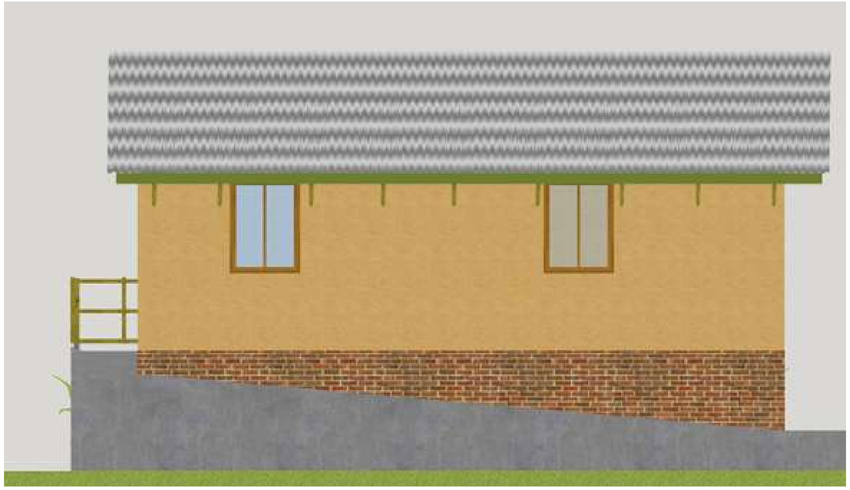
SCALE: NOT TO SCALE

DATE: **2079/80 AD**

SHEET NO. **A-03**

FRONT ELEVATION

SIDE ELEVATION



BACK ELEVATION

SIDE ELEVATION

DESIGNED BY:

CRS

PROJECT TITLE:
Flood Resilient Demo-house

DESIGN TYPE:
BAMBOO STRUCTURE WITH WATTLE & DAUB LIME STABILIZED SOIL INFILL

DRAWING TITLE:
ELEVATIONS

LOCATION:
Duduwa, Nepal

DRAWN BY:
CRS Nepal

N E S

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	900 X 2200	3
2.	WINDOW	W1	900 X 1200	4

AMENDABLE DETAILS

NON-AMENDABLE DETAILS

SCALE:
NOT TO SCALE

DATE:
2079/80 BS

SHEET NO:
A-04

FLOOD RESILIENT HOUSE TYPE DESIGN III:

RCC AND BAMBOO COMPOSITE CONSTRCUTION

INTRODUCTION

House Type: RCC and Bamboo Composite Construction

Designed By: Tukee- Nivas

Project Title: Shelter Program for Single Mothers (SPS)

Project Location:

1. Province: 2
2. District: Saptari
3. Ward: 9
4. Municipality: Rajbiraj

WHY IS THIS DESIGN FLOOD RESILIENT?

The design incorporates flood resilience through key features: an elevated plinth level to reduce flood impact, walls plastered for durability against water exposure, and a sturdy foundation reinforced with columns to enhance structural stability. Additionally, an attic is included, providing an emergency space during floods.

LOCATION MAP

CLIMATIC FEATURES:

Avg. Maximum Annual Temperature: 33.7˚C

Avg. Minimum Annual temperature: 12˚C

Annual Rainfall: 1270mm

Average relative Humidity: 85%

GEOTECHNICAL FEATURES:

Topography: Lowland Terrain

Mean Sea Level (MSL): 250-300 meters approx.

Soil Characteristics: Fertile Alluvial soil

Disaster: Seasonal flooding

RCC AND BAMBOO COMPOSITE CONSTRCUTION

DESIGN CONSIDERATIONS

Available Building Materials: Brick, Cement, Sand, Aggregate, Rebar, CGI Sheet, Mud

Foundation: Brick Masonry with Cement Mortar

Plinth: PPC

Fence/Wall: Masonry Wall

Openings: Timber frame, Plywood Shutter

Ceiling: CGI Sheet

Joints: N/A

Treatment (bamboo & wood): N/A

Roof Type: Pitch Roof

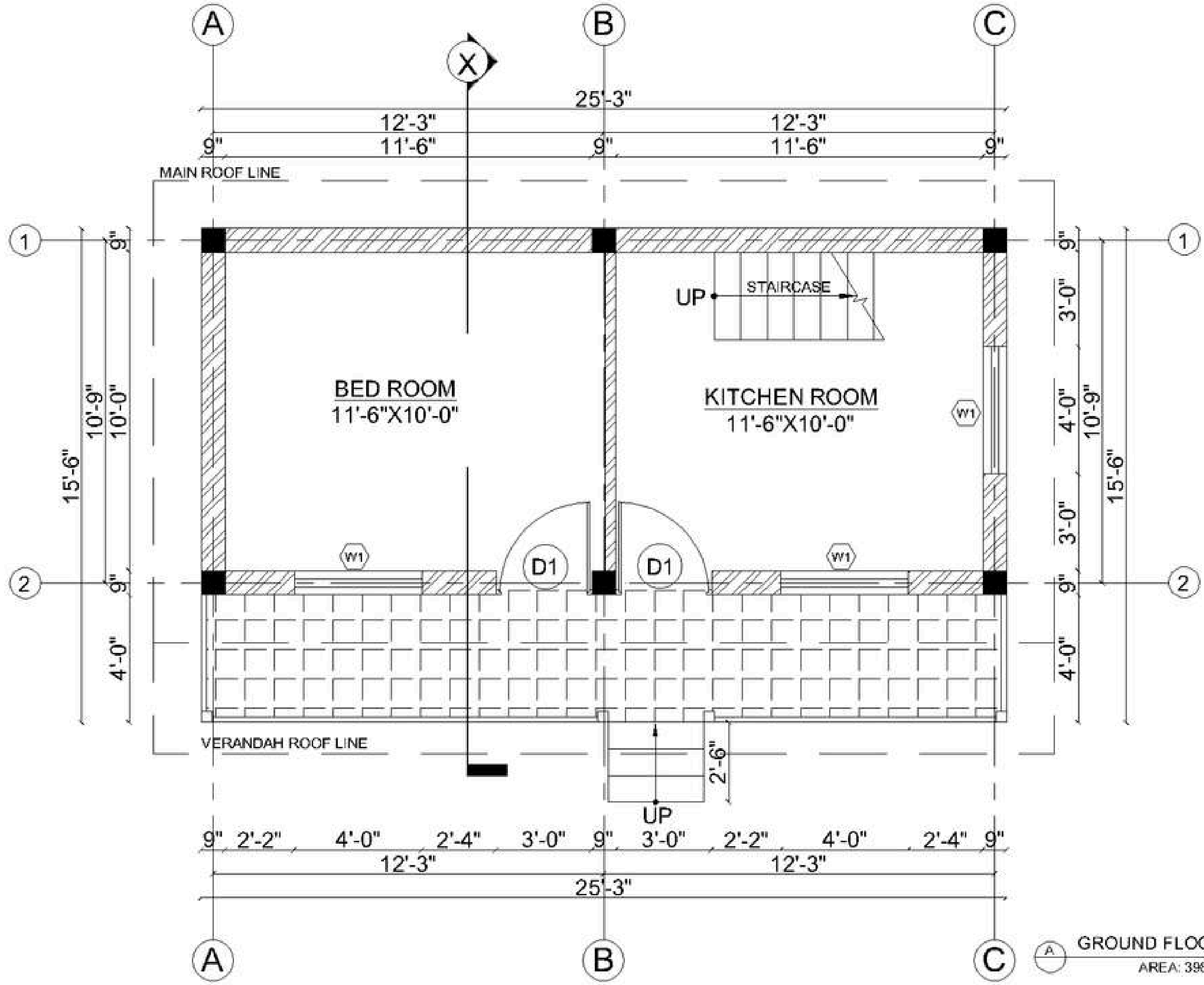
Roof cover: CGI Sheet

Roof structure: Black Pipe with J-Hook for Purlins

Bracing: N/A

Note: Rate analysis based on field experience

LEVEL	MANPOWER		MATERIAL										
	SKILLED	UNSKILLED	BRICK	CEMENT	SAND	AGGREGATE	WOOD			CGI SHEET	PLAIN SHEET	SQUARE MS PIPE	REINFORCEMENT BARS
							SAL	LOCAL	BAMBOO				
	Md	Md	No.s	Bags	Cu.m	Cu.m	Cu.m	Sq.m		Bundle	running meter	KG	KG
UP TO PLINTH LVL	38	95	7129	12	33.64	3.7	-	-	-	-	-	-	-
SUPERSTRUCTURE	149	153	5927	116	11.18	6.07	0.69	0.35	45	-	-	-	1165
ROOFING	29	33	-	-	-	-	-	-	-	11.38	9.88	588	-
TOTAL	**216**	**281**	**13056**	**128**	**44.82**	**9.77**	**0.69**	**0.35**	**45**	**11.38**	**9.88**	**588**	**1165**



A GROUND FLOOR PLAN

AREA: 398.87 SQ.FT

DESIGNED BY:

PROJECT TITLE:
SHELTER PROGRAM FOR SINGLE MOTHERS (SPS) PROJECT

DESIGN TYPE:
RCC AND BAMBOO COMPOSITE CONSTRUCTION

DRAWING TITLE:
FLOOR PLANS

LOCATION:
RAJBIRAJ, NEPAL

DRAWN BY:
TUKEE, NIVAS

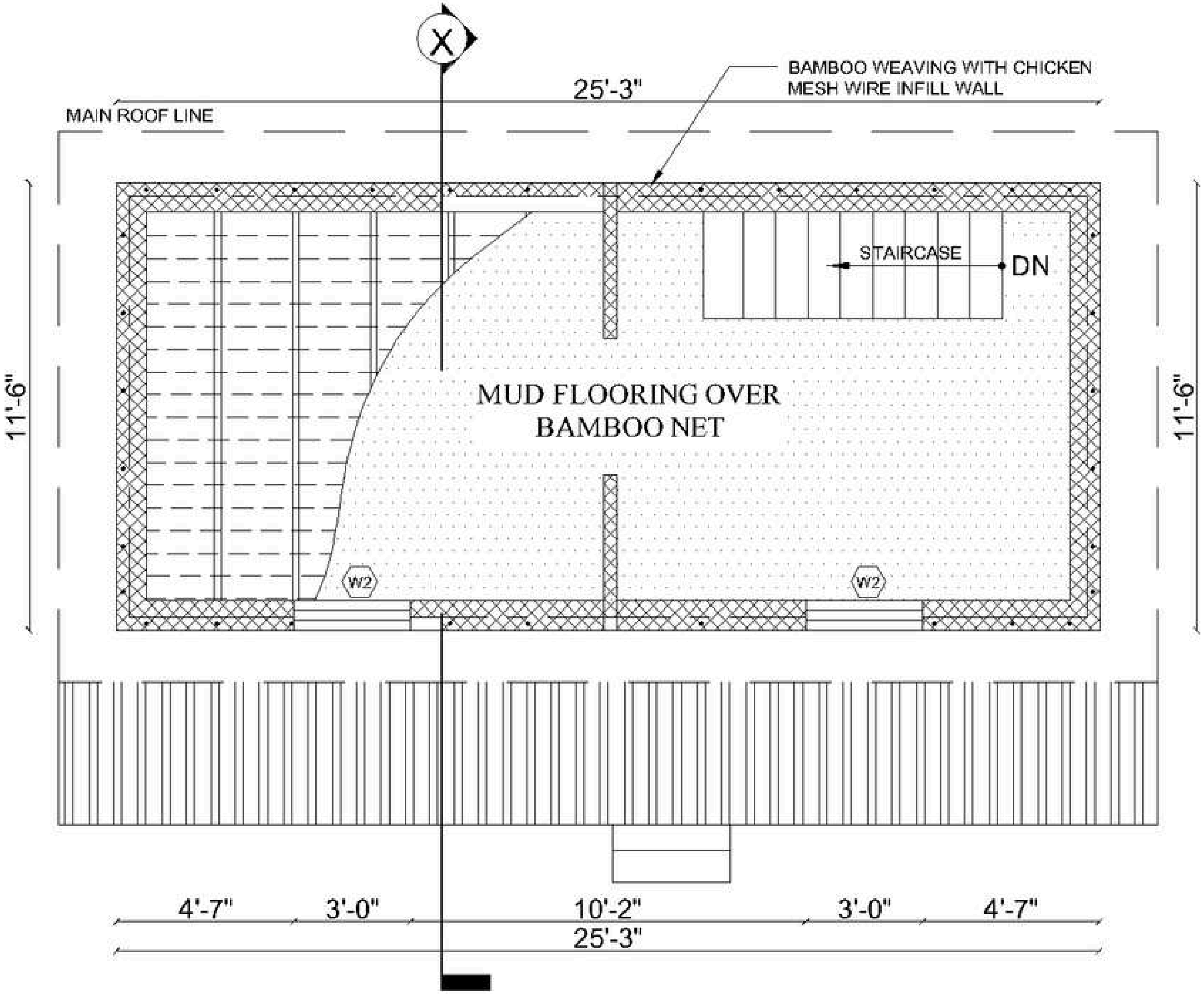
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7'-0"	2
1.	DOOR	D2	2'-0" X 7'-0"	2
2.	WINDOW	W1	4'-0" X 4'-6"	1
3.	WINDOW	W2	3'-0" X 1'-6"	3

SCALE:
NOT TO SCALE

SHEET NO.
A-01

DATE:
2022 AD



A

ATTIC FLOOR PLAN

AREA: 398.87 SQ.FT

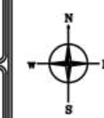
DESIGNED BY:

PROJECT TITLE:
SHELTER PROGRAM FOR SINGLE MOTHERS (SPS) PROJECT

DESIGN TYPE:
RCC AND BAMBOO COMPOSITE CONSTRUCTION

DRAWING TITLE:
FLOOR PLANS

LOCATION:
RAJBIRAJ, NEPAL



DRAWN BY:
TUKEE, NIVAS

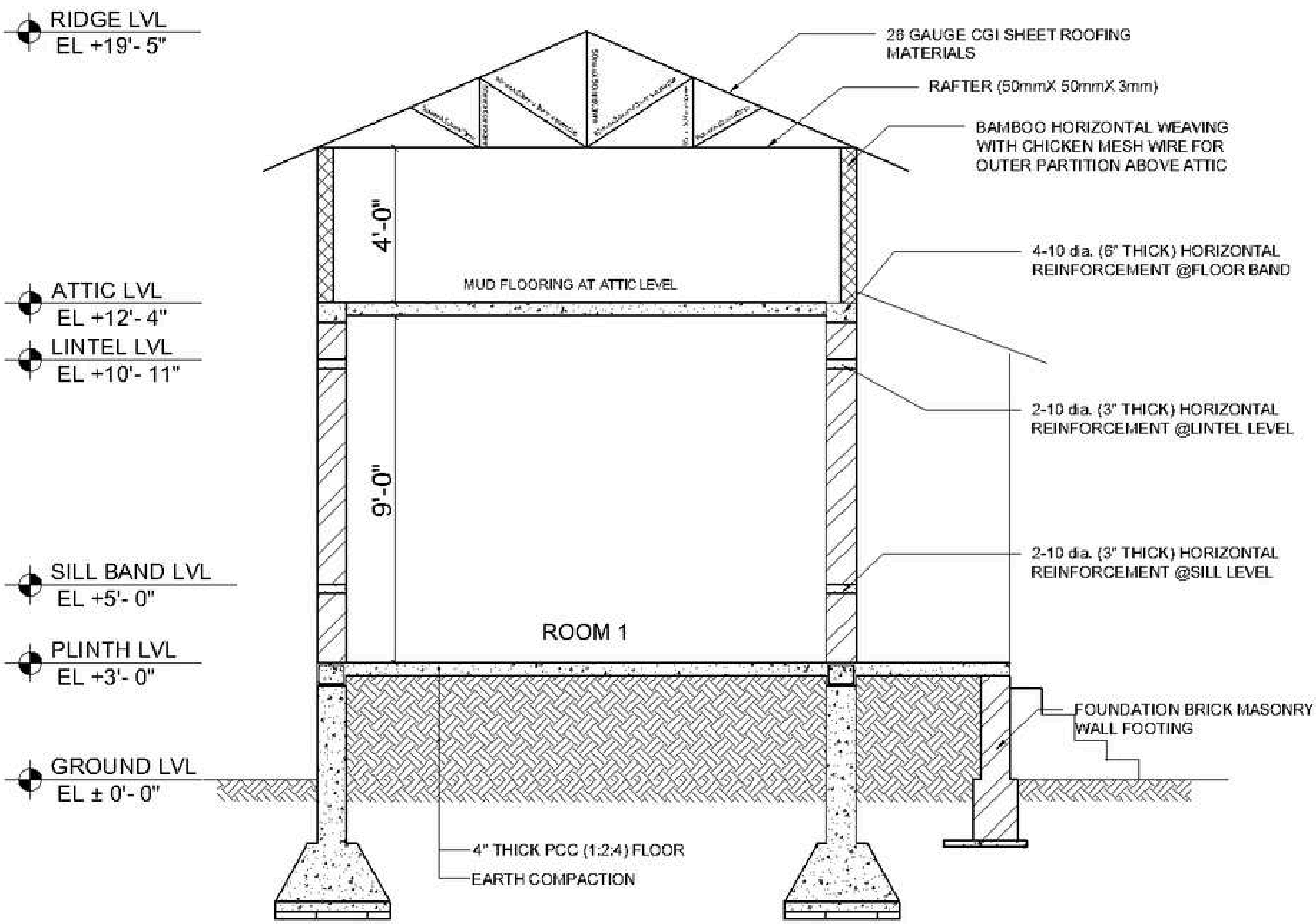
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7'-0"	2
1.	DOOR	D2	2'-0" X 7'-0"	2
2.	WINDOW	W1	4'-0" X 4'-6"	1
3.	WINDOW	W2	3'-0" X 1'-6"	3

SCALE:
NOT TO SCALE

SHEET NO.
A-02

DATE:
2022 AD



SECTION AT X-X

A SECTION AT X-X

DESIGNED BY:

PROJECT TITLE:
SHELTER PROGRAM FOR SINGLE MOTHERS (SPS) PROJECT

DESIGN TYPE:
RCC AND BAMBOO COMPOSITE CONSTRUCTION

DRAWING TITLE:
SECTIONAL DETAILS

LOCATION:
RAJBIRAJ, NEPAL

DRAWN BY:
TUKEE, NIVAS

OPENING SCHEDULE

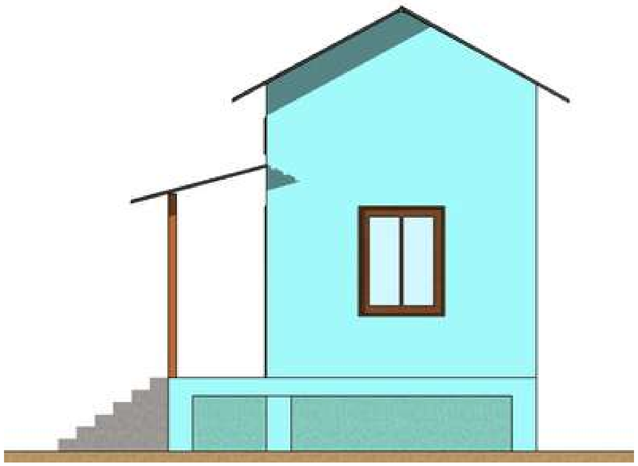
S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7'-0"	2
1.	DOOR	D2	2'-0" X 7'-0"	2
2.	WINDOW	W1	4'-0" X 4'-6"	1
3.	WINDOW	W2	3'-0" X 1'-6"	3

SCALE:
NOT TO SCALE

DATE:
2022 AD

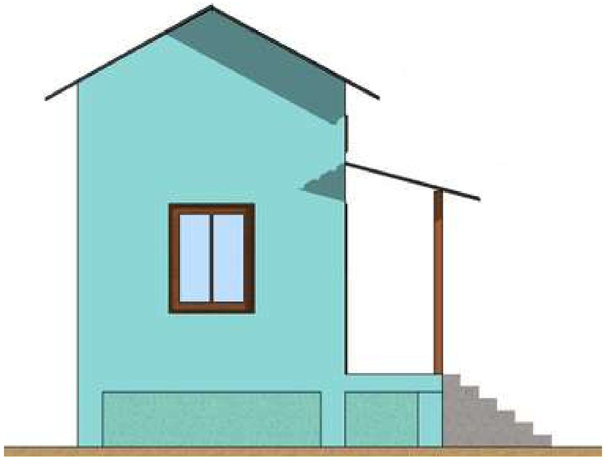
SHEET NO.
A-03

FRONT ELEVATION



SIDE ELEVATION

BACK ELEVATION



SIDE ELEVATION

DESIGNED BY:

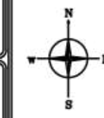
PROJECT TITLE:
SHELTER PROGRAM FOR SINGLE MOTHERS (SPS) PROJECT

DESIGN TYPE:
RCC AND BAMBOO COMPOSITE CONSTRUCTION

DRAWING TITLE:
ELEVATIONS

LOCATION:
RAJBIRAJ, NEPAL

DRAWN BY:
TUKEE, NIVAS



OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7'-0"	2
1.	DOOR	D2	2'-0" X 7'-0"	2
2.	WINDOW	W1	4'-0" X 4'-6"	1
3.	WINDOW	W2	3'-0" X 1'-6"	3

SCALE:
NOT TO SCALE

DATE:
2022 AD

SHEET NO.
A-04

FLOOD RESILIENT HOUSE TYPE DESIGN IV:

COMPRESSED STABILIZED EARTH BRICKS CONSTRUCTION

INTRODUCTION

House Type: Compressed Stabilized Earth Bricks (CSEB) Construction
Designed By: Build-Up Nepal
Project Title: Pratappur Housing Project
Project Location:

1. Province: Lumbini
2. District: Nawalparasi
3. Municipality: Pratappur

LOCATION MAP

CLIMATIC FEATURES:
Avg. Maximum Annual Temperature: 25-30˚C
Avg. Minimum Annual temperature: 10-15˚C
Annual Rainfall: Above1500mm
Average relative Humidity: Above 80%

GEOTECHNICAL FEATURES:
Topography: Hilly and mountainous
Mean Sea Level (MSL): 800- 2500 metres
Soil Characteristics: Loose and porous
Disaster: Landslides and flooding

WHY IS THIS DESIGN FLOOD RESILIENT?

- **Durable Foundation Bricks:** Made from stone dust and cement, the bricks are resistant to decomposition, even in prolonged waterlogged conditions, ensuring a strong and stable foundation.
- **Water Resilience:** The materials used enhance the durability of the structure, maintaining integrity during flooding.
- **Elevated Plinth Level:** The plinth level is raised to keep rooms safe from water intrusion during floods, adding an essential layer of protection

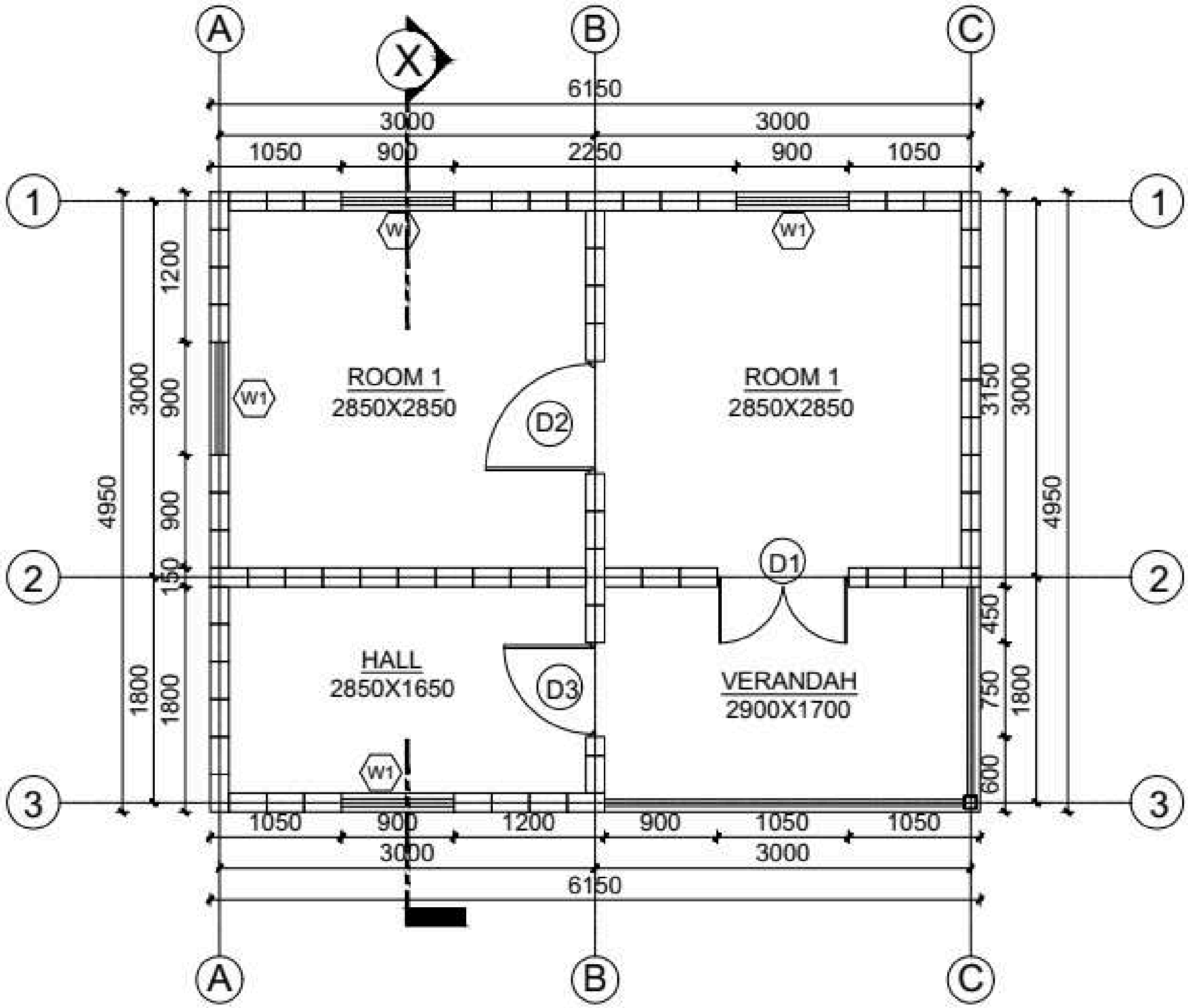
COMPRESSED STABILIZED EARTH BRCIKS CONSTRUCTION

DESIGN CONSIDERATIONS

Available Building Materials: Stone, Cement, Wood, Earth, Sand, Aggregate
Foundation: Stone Masonry in Cement Mortar
Plinth: RCC
Post: Mild Steel Tubular Post
Fence/Wall: CSEB/ Stone Dust Interlocking Block
Openings: Wood, Aluminum, MS
Ceiling: Plywood/ Gypsum
Joints: N/A
Treatment (bamboo & wood): N/A
Roof Type: Pitch Roof
Roof cover: CGI/UPVC/Slate
Roof structure: MS Truss/ Wooden
Bracing: N/A

Note: Rate analysis based on field experience

LEVEL	MANPOWER		MATERIAL								
	SKILLED	UNSKILLED	STONE	INTERLOCKING BLOCKS	CEMENT	SAND	AGGREGATE	METAL TRUSS	CGI SHEET	REFORCEMENT BARS	
										10MM	8MM
	Md	Md	Cu.ft	No.s	Bags	Cu.ft	Cu.ft	KG	Bundle	Kg	Kg
UP TO PLINTH LVL	20	78	1401	-	51	198	168	-	-	224	59
SUPERSTRUCTURE	18	53	-	2203	21	89	52	-	-	148	99
ROOFING	-	-	-	-	-	-	-	358	4	-	-
TOTAL	**38**	**131**	**1401**	**2203**	**72**	**287**	**220**	**358**	**4**	**372**	**158**



A GROUND FLOOR PLAN
AREA: 30 SQ.M

DESIGNED BY:

BUILD UP NEPAL

PROJECT TITLE:
PRATAPPUR HOUSING PROJECT

DESIGN TYPE:
COMPRESSED STABILIZED EARTH BRICK CONSTRUCTION

DRAWING TITLE:
FLOOR PLANS

LOCATION:
PRATAPPUR, NEPAL

DRAWN BY:
BUILD-UP NEPAL

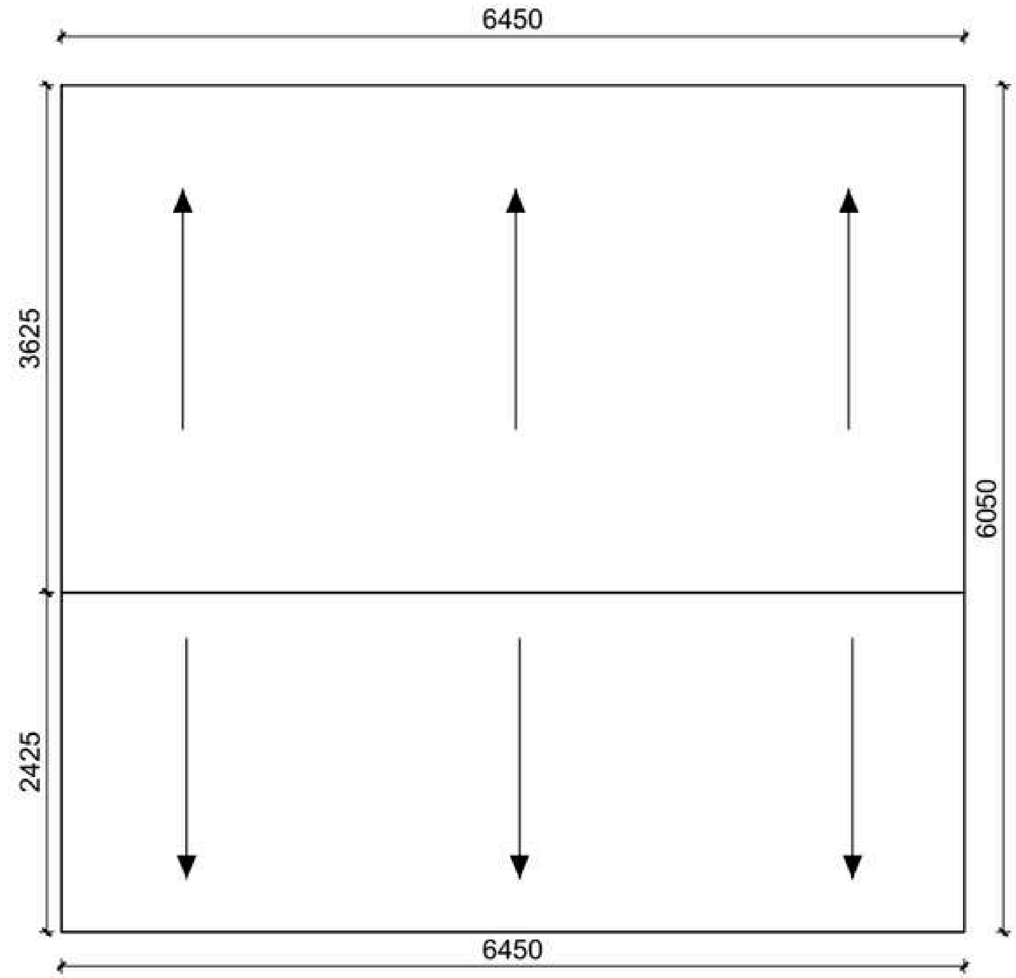
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	1050 X 2100	4
2.	DOOR	D2	900 X 2100	4
3.	DOOR	D3	750 X 2100	4
4.	WINDOW	W1	900 X 1205	5

SCALE:
NOT TO SCALE

DATE:
2017 AD

SHEET NO.
A-01



ROOF PLAN
A
1:50

DESIGNED BY:

BUILD UP NEPAL

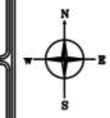
PROJECT TITLE:
PRATAPPUR HOUSING PROJECT

DESIGN TYPE:
COMPRESSED STABILIZED EARTH BRICK CONSTRUCTION

DRAWING TITLE:
ROOF PLAN

LOCATION:
PRATAPPUR, NEPAL

DRAWN BY:
BUILD-UP NEPAL



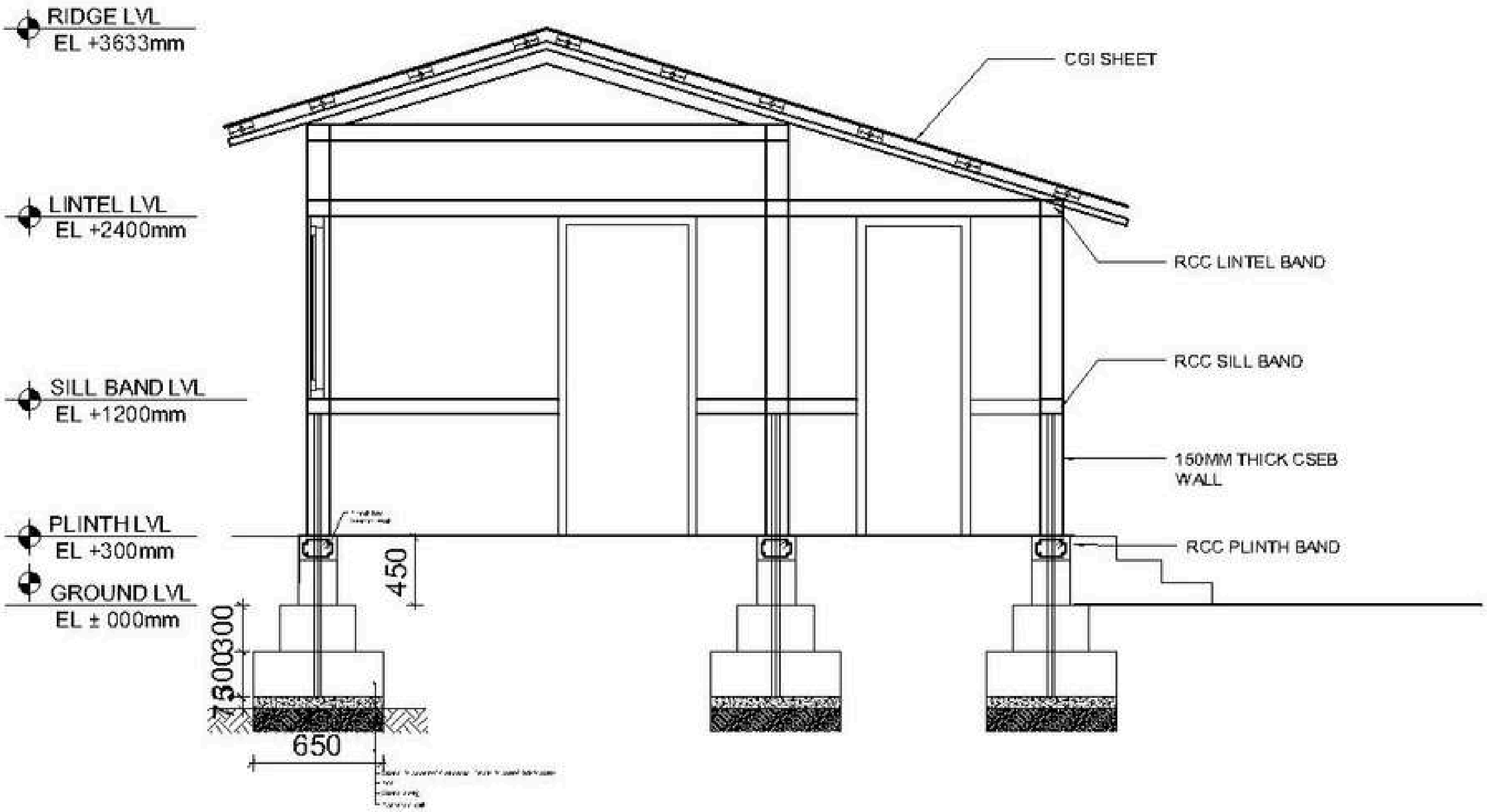
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	1050 X 2100	4
2.	DOOR	D2	900 X 2100	4
3.	DOOR	D3	750 X 2100	4
4.	WINDOW	W1	900 X 1205	5

SCALE:
NOT TO SCALE

DATE:
2017 AD

SHEET NO.
A-02



A SECTION AT X-X
SCALE: 1:50

DESIGNED BY:

BUILD UP NEPAL

PROJECT TITLE:
PRATAPPUR HOUSING PROJECT

DESIGN TYPE:
COMPRESSED STABILIZED EARTH BRICK CONSTRUCTION

DRAWING TITLE:
SECTIONAL DETAILS

LOCATION:
PRATAPPUR, NEPAL

DRAWN BY:
BUILD-UP NEPAL

N W E S

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	1050 X 2100	4
2.	DOOR	D2	900 X 2100	4
3.	DOOR	D3	750 X 2100	4
4.	WINDOW	W1	900 X 1205	5

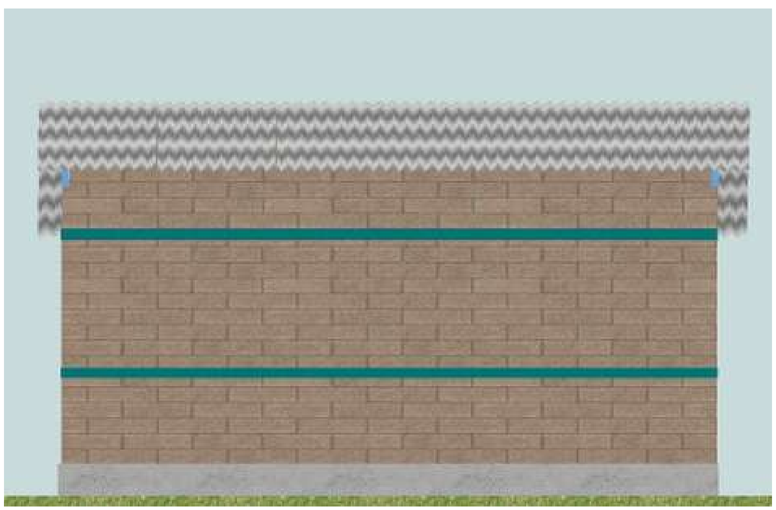
SCALE:
NOT TO SCALE

DATE:
2017 AD

SHEET NO.
A-03

FRONT ELEVATION

SIDE ELEVATION



BACK ELEVATION

SIDE ELEVATION

DESIGNED BY:

BUILD UP NEPAL

PROJECT TITLE:
PRATAPPUR HOUSING PROJECT

DESIGN TYPE:
COMPRESSED STABILIZED EARTH BRICK CONSTRUCTION

DRAWING TITLE:
ELEVATIONS

LOCATION:
PRATAPPUR, NEPAL

DRAWN BY:
BUILD-UP NEPAL

N W E S

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	1050 X 2100	4
2.	DOOR	D2	900 X 2100	4
3.	DOOR	D3	750 X 2100	4
4.	WINDOW	W1	900 X 1205	5

SCALE:
NOT TO SCALE

DATE:
2017 AD

SHEET NO.
A-04

FLOOD RESILIENT HOUSE TYPE DESIGN V:

BRICK MASONARY WITH CEMENT MORTAR

INTRODUCTION

House Type: Brick Masonry with Cement Mortar

Designed By: Caritas Nepal

Project Title: Low-cost Housing Construction

Project Location:

1. Province: Lumbini
2. District: Bardiya
3. Municipality: Gulariya

LOCATION MAP

CLIMATIC FEATURES:

Avg. Maximum Annual Temperature: 36.8˚C

Avg. Minimum Annual temperature: 8˚C

Annual Rainfall: 1245 mm

Average relative Humidity: 87%

GEOTECHNICAL FEATURES:

Topography: Flat Terai Plains

Mean Sea Level (MSL): 195 meters approx.

Soil Characteristics: Fertile Alluvial Soil

Disaster: Seasonal Flooding

WHY IS THIS DESIGN FLOOD RESILIENT?

Flood Resilient features includes:

1. Raised plinth level
2. Structurally strong brick walls with cement mortar for flood resilience
3. Strong connections following DUDBC norms - vertical ties , roof band, corner bands, plinth band, sill band
4. Wind resistance through better connection detailing of metal truss and Corrugated Galvanized Iron CGI sheets

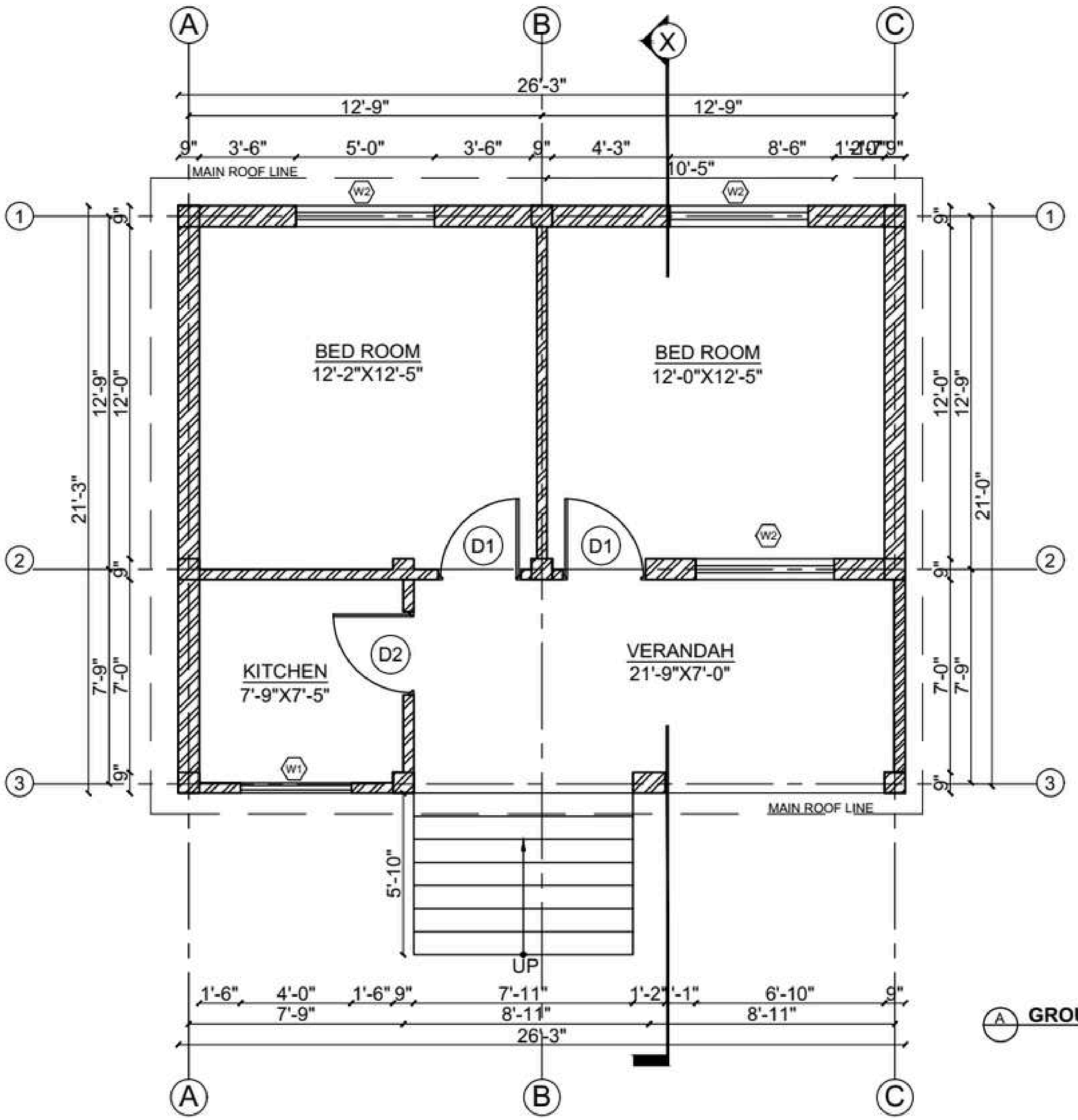
BRICK MASONRY WITH CEMENT MORTAR

DESIGN CONSIDERATIONS

Available Building Materials: Brick, Cement, Sand, Aggregate, Rebar, CGI Sheet, Mud
Foundation: Brick Masonry with Cement Mortar
Plinth: PPC
Fence/Wall: Masonry Wall
Openings: Timber frame, Plywood Shutter
Ceiling: CGI Sheet
Joints: N/A
Treatment (bamboo & wood): N/A
Roof Type: Pitch Roof
Roof cover: CGI Sheet
Roof structure: Black Pipe with J-Hook for Purlins
Bracing: N/A

Note: Rate analysis based on field experience

LEVEL	MANPOWER		MATERIAL							
	SKILLED	UNSKILLED	STONE	BRICK	CEMENT	SAND	AGGREGATE	WOOD	CGI SHEET	REFORCEMENT BARS
	Md	Md	Cu.m	No.s	Bags	Cu.m	Cu.m	Cu.m	Bundle	Kg
UP TO PLINTH LVL	44	104	39.67	13115	81	11	7	-	-	146
SUPERSTRUCTURE	67	59	-	8984	46	5	2	0.79	-	314
ROOFING	17	20	-	-	-	-	-	1.48	4.71	-
TOTAL	**129**	**183**	**39.67**	**22099**	**127**	**16**	**9**	**2.27**	**4.71**	**460**



A GROUND FLOOR PLAN

AREA: 557.81 SQ.FT

DESIGNED BY:

PROJECT TITLE:
LOW COST HOUSING CONSTRUCTION

DESIGN TYPE:
BRICK MASONRY WITH CEMENT MORTAR

DRAWING TITLE:
FLOOR PLANS

LOCATION:
GULARIYA NEPAL

DRAWN BY:
CARITAS NEPAL

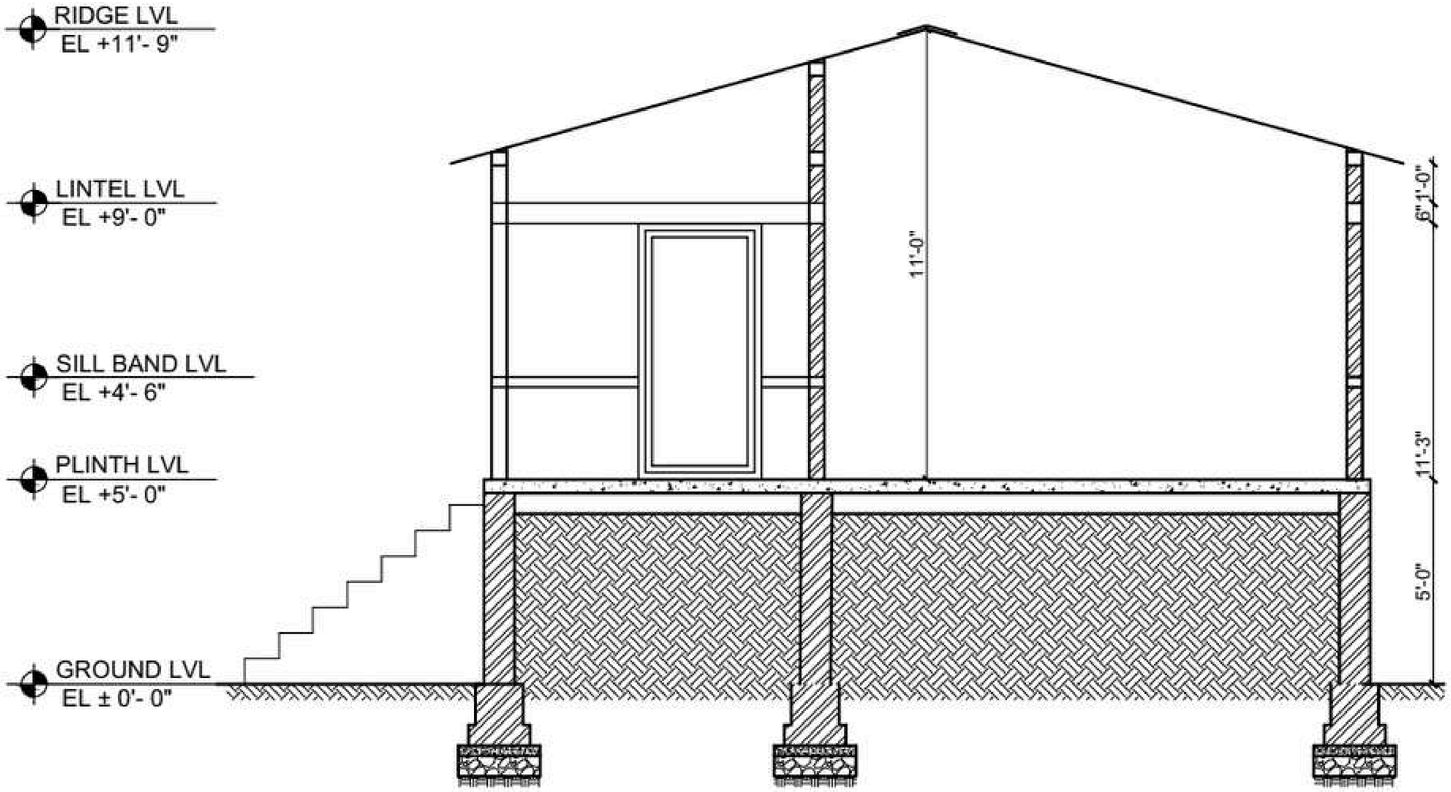
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 6'-0"	2
2.	DOOR	D2	3'-0" X 6'-0"	1
3.	WINDOW	W1	4'-0" X 4'-0"	1
4.	WINDOW	W2	5'-0" X 4'-0"	3

SCALE: NOT TO SCALE

DATE: **2079/80 BS**

SHEET NO. **A-01**



A

SECTION AT X-X
SCALE: 1:40

DESIGNED BY:

PROJECT TITLE:
LOW COST HOUSING CONSTRUCTION

DESIGN TYPE:
BRICK MASONRY WITH CEMENT MORTAR

DRAWING TITLE:
SECTIONAL DETAILS

LOCATION:
GULARIYA NEPAL

DRAWN BY:
CARITAS NEPAL

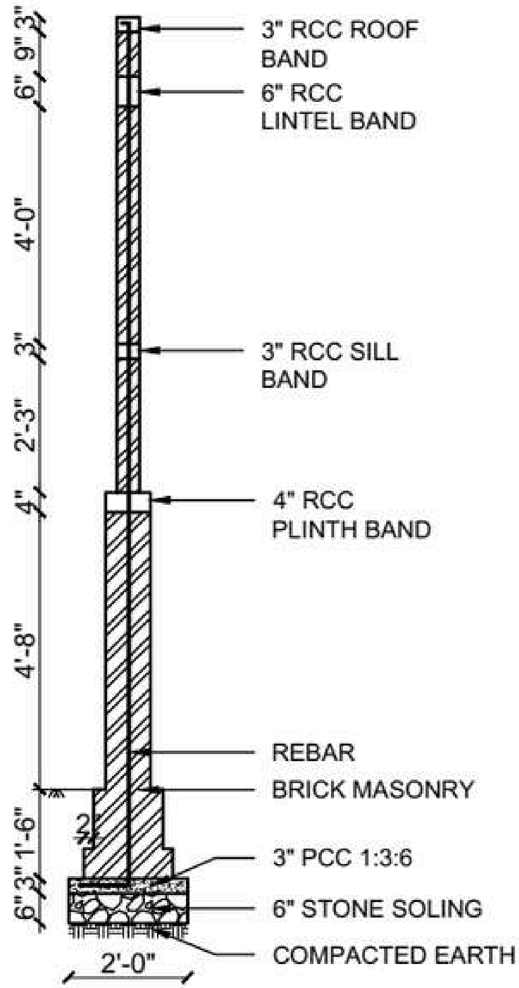
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 6'-0"	2
2.	DOOR	D2	3'-0" X 6'-0"	1
3.	WINDOW	W1	4'-0" X 4'-0"	1
4.	WINDOW	W2	5'-0" X 4'-0"	3

SCALE:
NOT TO SCALE

SHEET NO.
A-02

DATE:
2079/80 BS



A WALL SECTION
SCALE: 1:30

DESIGNED BY:

PROJECT TITLE:
LOW COST HOUSING CONSTRUCTION

DESIGN TYPE:
BRICK MASONRY WITH CEMENT MORTAR

DRAWING TITLE:
SECTIONAL DETAILS

LOCATION:
GULARIYA NEPAL

DRAWN BY:
CARITAS NEPAL

OPENING SCHEDULE

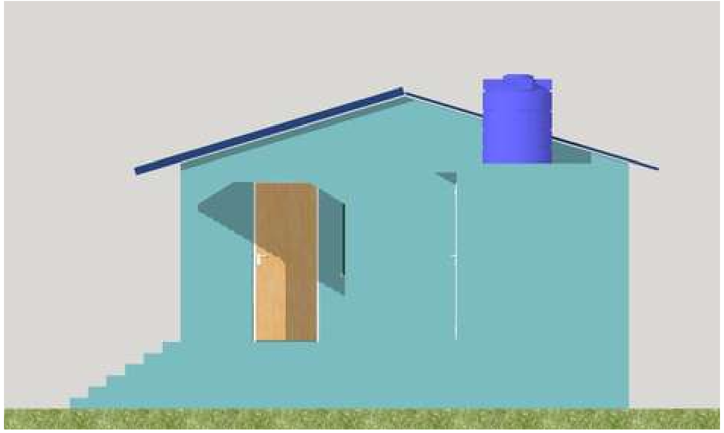
S.N	NAME	CODE	SIZE	NOs
1.	DOOR	D1	3'-0" X 6'-0"	2
2.	DOOR	D2	3'-0" X 6'-0"	1
3.	WINDOW	W1	4'-0" X 4'-0"	1
4.	WINDOW	W2	5'-0" X 4'-0"	3

SCALE: NOT TO SCALE

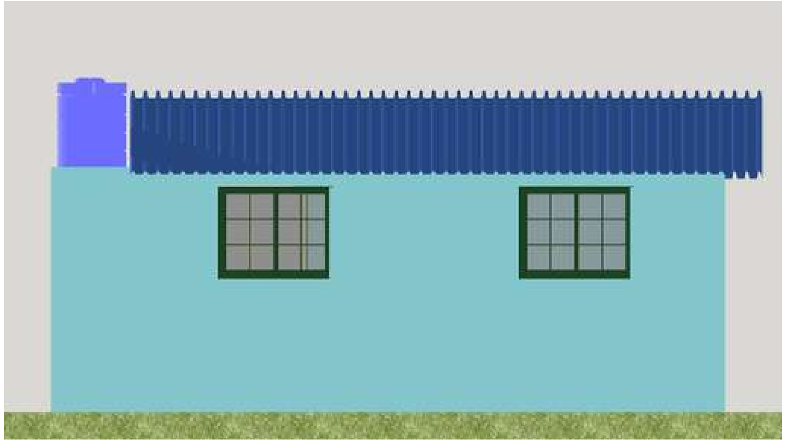
DATE: **2079/80 BS**

SHEET NO. **A-03**

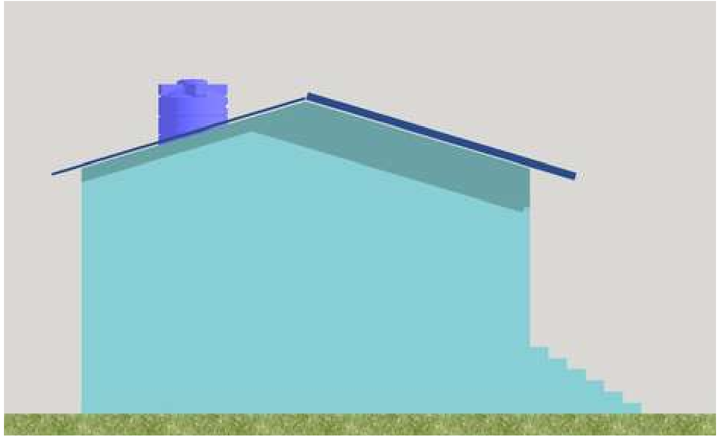
FRONT ELEVATION



SIDE ELEVATION



BACK ELEVATION



SIDE ELEVATION

DESIGNED BY:

PROJECT TITLE:
LOW COST HOUSING CONSTRUCTION

DESIGN TYPE:
BRICK MASONRY WITH CEMENT MORTAR

DRAWING TITLE:
ELEVATIONS

LOCATION:
GULARIYA NEPAL

DRAWN BY:
CARITAS NEPAL

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 6'-0"	2
2.	DOOR	D2	3'-0" X 6'-0"	1
3.	WINDOW	W1	4'-0" X 4'-0"	1
4.	WINDOW	W2	5'-0" X 4'-0"	3

SCALE:
NOT TO SCALE

DATE:
2079/80 BS

SHEET NO.
A-04

FLOOD RESILIENT HOUSE TYPE DESIGN VI:

DOME STRUCTURE RAISED ON METAL STILTS

INTRODUCTION

House Type: Dome Structure raised on Metal Stilts
Designed By: Chitwan Architecture Design Studio (CADS) Pvt.Ltd
Structure Analysis by: Kamero Consults Pvt.Ltd
Project Title: Nepal Himalayan Paulowina Resort
Project Location:

1. Province: Bagmati
2. District: Chitwan
3. Municipality: Bharatpur Metropolitan

LOCATION MAP

CLIMATIC FEATURES:
Avg. Maximum Annual Temperature: 35-40˚C
Avg. Minimum Annual temperature: 5˚C
Annual Rainfall: 2000-2500 mm
Average relative Humidity: 80-90%

GEOTECHNICAL FEATURES:
Topography: Flat Terai Plains
Mean Sea Level (MSL): 250-300 meters approx.
Soil Characteristics: Fertile Alluvial Soil
Disaster: Seasonal Flooding

WHY IS THIS DESIGN FLOOD RESILIENT?

he structure is elevated 4 feet above the ground on sturdy metal pilotis. This elevated position ensures that in the event of flooding, water and debris can flow freely beneath the house without causing damage or obstruction.

Raised design provides a barrier against land reptiles for safety and comfort of the occupants. Open space beneath the structure is utilized efficiently during dry periods, serving as a grazing area for chickens. This dual-purpose approach not only maximizes land use but also supports sustainable practices, blending practicality with environmental adaptability.

In 2023, highest recorded water level in the area reached 3 feet, well below the elevated platform, keeping the structure safe during floods.

CHITWAN ARCHITECTURE DESIGN STUDIO

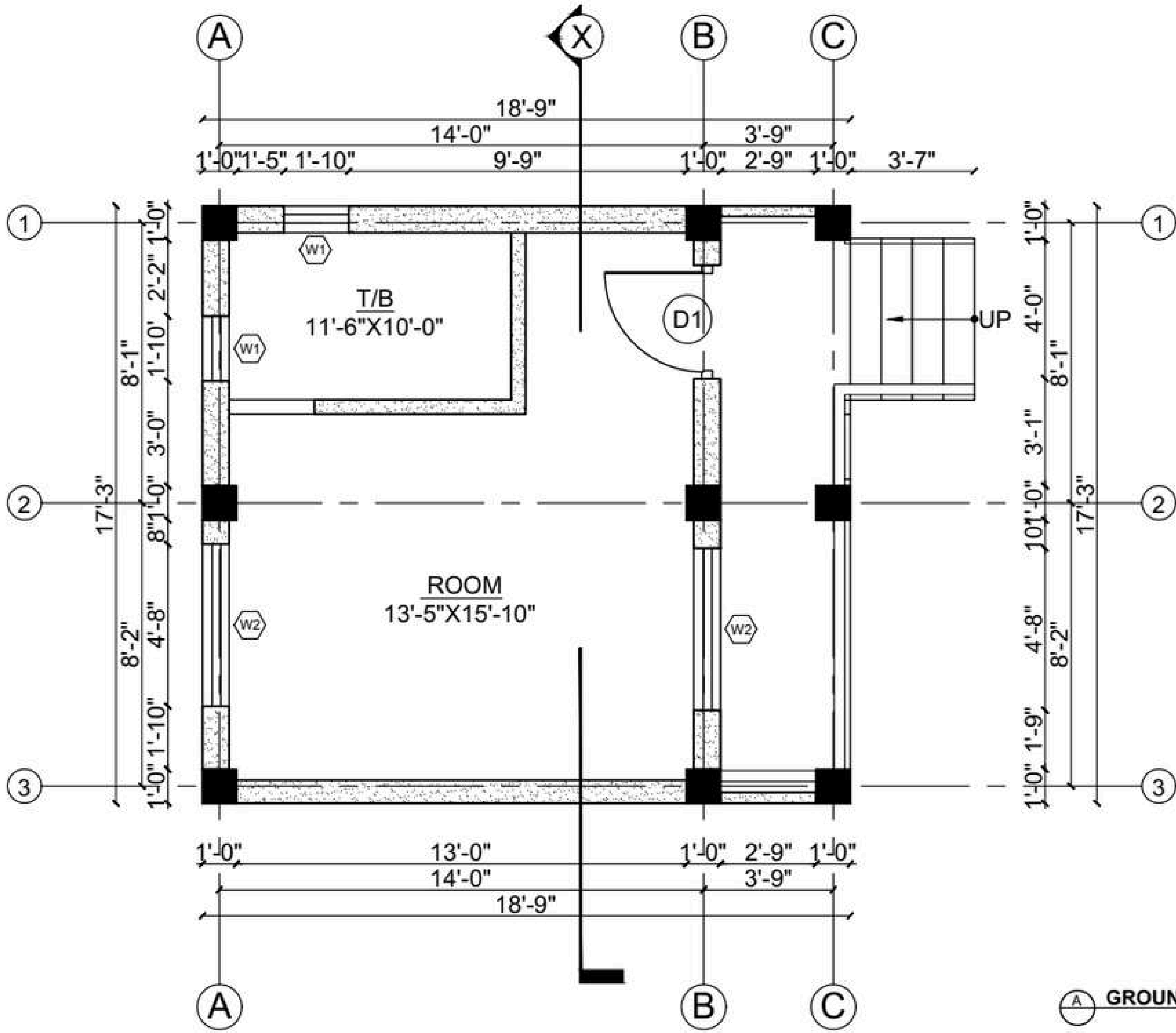
DOME STRUCTURE RAISED ON METAL STILTS

DESIGN CONSIDERATIONS

Available Building Materials: Stone, Metal, GI sheet, cement, red mud, Bamboo, Straw
Foundation: Isolated RCC footing
Plinth: 4' raised plinth, metal structure
Fence/Wall: Metal covered with Ferrocement
Openings: Bamboo, wood
Ceiling: Bamboo
Treatment (bamboo & wood): Borax treatment
Roof Type: Two way dome
Roof cover: GI sheets, thatch
Roof structure: Dome
Bracing: Metal cross bracings on elevated plinths
Joints: Knot bolt, Welding

Note: Rate analysis based on field experience

LEVEL	MANPOWER		MATERIAL						
	SKILLED	UNSKILLED	STONE	MUD	CEMENT	SAND	AGGREGATE	BAMBOO	STEEL REFORCEMENT BARS
	Md	Md	Cu.ft	Cu.ft	Bags	Cu.m	Cu.ft	No.s	Kg
UP TO PLINTH LVL	45	60	78.8	-	30	11	84.88	-	547.5
SUPERSTRUCTURE	60	30	-	32.13	30	5	-	130	3181
TOTAL	**105**	**183**	**78.8**	**32.13**	**60**	**16**	**84.88**	**130**	**3728.5**



A **GROUND FLOOR PLAN**
AREA: 557.81 SQ.FT

DESIGNED BY:

CHITWAN ARCHITECTURE DESIGN STUDIO

PROJECT TITLE:
NEPAL HIMALAYAN PAULOWINA RESORT

DESIGN TYPE:
DOME STRUCTURE RAISED ON METAL STILTS

DRAWING TITLE:
FLOOR PLANS

LOCATION:
MEGHAULI, NEPAL

DRAWN BY: **CADS**

STRUCTURE ANALYSIS:
KAMERO CONSULTS

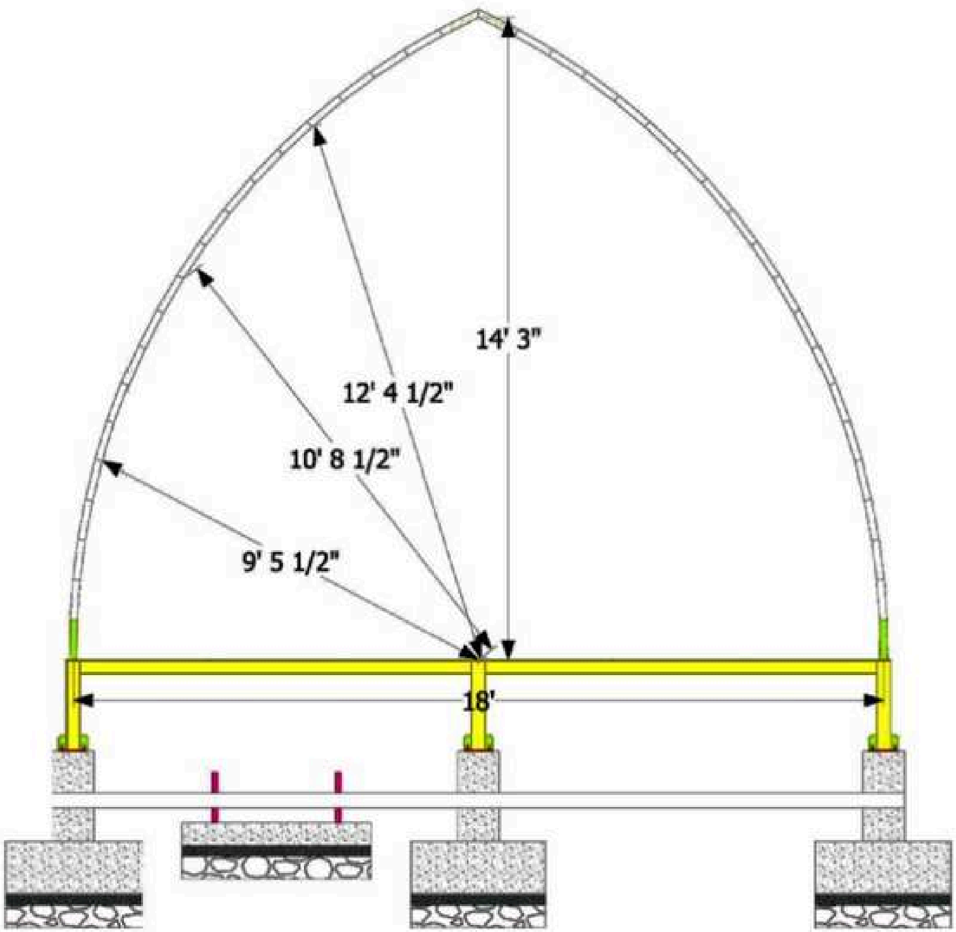
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 6'-0"	1
2.	WINDOW	W1	1'-10"X 2'-2"	2
3.	WINDOW	W2	4'-8" X 4'-6"	2

SCALE:
NOT TO SCALE

DATE:
2021 AD

SHEET NO.
A-01



SECTION AT X-X
NOT TO SCALE

DESIGNED BY:

CHITWAN ARCHITECTURE DESIGN STUDIO

PROJECT TITLE:
NEPAL HIMALAYAN PAULOWINA RESORT

DESIGN TYPE:
DOME STRUCTURE RAISED ON METAL STILTS

DRAWING TITLE:
SECTIONAL DETAILS

LOCATION:
MEGHAULI, NEPAL

DRAWN BY: **CADS**

STRUCTURE ANALYSIS:
KAMERO CONSULTS

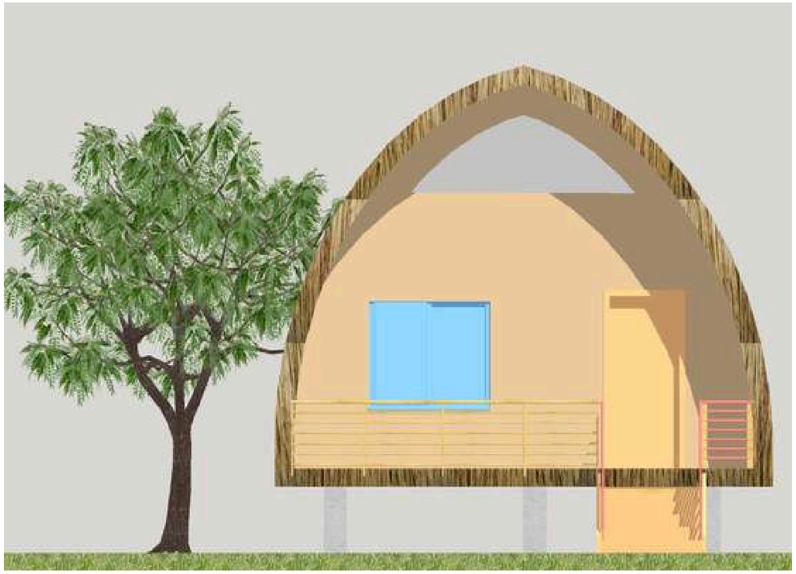
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 6'-0"	1
2.	WINDOW	W1	1'-10"X 2'-2"	2
3.	WINDOW	W2	4'-8" X 4'-6"	2

SCALE:
NOT TO SCALE

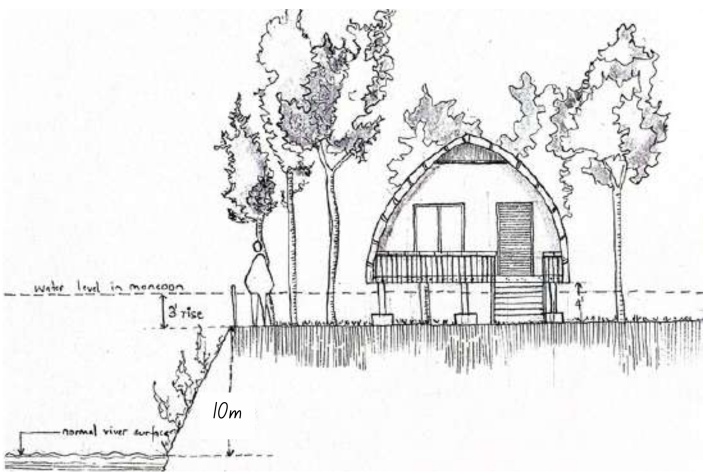
DATE:
2021 AD

SHEET NO.
A-02

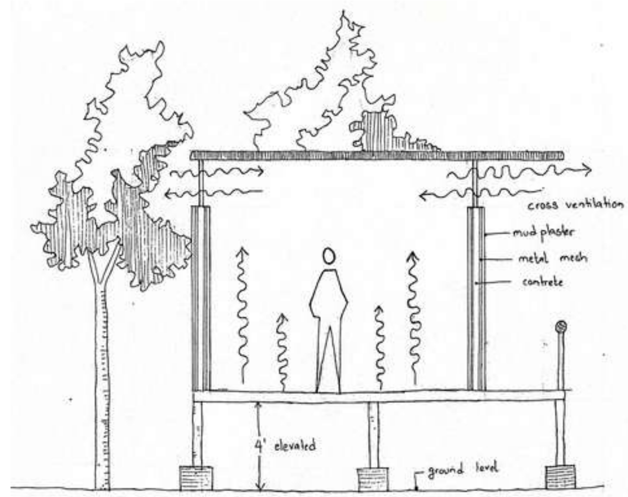


FRONT ELEVATION

SIDE ELEVATION



ELEVATION SKETCH



SECTION ELEVATION

DESIGNED BY:

CHITWAN ARCHITECTURE DESIGN STUDIO

PROJECT TITLE:
NEPAL HIMALAYAN PAULOWINA RESORT

DESIGN TYPE:
DOME STRUCTURE RAISED ON METAL STILTS

DRAWING TITLE:
ELEVATIONS

LOCATION:
MEGHAULI, NEPAL

N S W E

DRAWN BY: **CADS**

STRUCTURE ANALYSIS:
KAMERO CONSULTS

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 6'-0"	1
2.	WINDOW	W1	1'-10"X 2'-2"	2
3.	WINDOW	W2	4'-8" X 4'-6"	2

SCALE:
NOT TO SCALE

SHEET NO.
A-03

DATE:
2021 AD

FLOOD RESILIENT HOUSE TYPE DESIGN VII:

RAT-TRAP BOND MASONRY WALL CONSTRUCTION

INTRODUCTION

House Type: Rat-trap Bond Masonry Wall Construction

Designed By: Habitat for Humanity

Project Title: Community Housing Project

Project Location:

1. Province: Sudurpaschim
2. District: Kanchanpur

LOCATION MAP

CLIMATIC FEATURES:

Avg. Maximum Annual Temperature: 30.82˚C

Avg. Minimum Annual temperature: 18.2˚C

Annual Rainfall: Varies, Data from 2016-2022 shows, lowest- 993.7mm in 2019 and highest- 1990.6 in 2016

Average relative Humidity: At 8:45am: 82.64%, At 5:45pm: 68.04%

(Data from nearby weather station)

GEOTECHNICAL FEATURES:

Topography: Flat Terai Plains

Mean Sea Level (MSL): 184 meters approx.

Soil Characteristics: Fertile Alluvial Soil

Disaster: Seasonal Inundation

WHY IS THIS DESIGN FLOOD RESILIENT?

- Construction site is selected on comparatively elevated land with low risk of floods and inundation.
- Materials that are less affected / unaffected by water are selected.
- Sufficient projection of roof overhangs are provided to protect the walls from rain and increase the durability of materials.
- Roads are designed with drain lines of sufficient sizes for storm water to pass.

The house is performing well in rain providing safe shelter to the residents.

Habitat for Humanity® Nepal

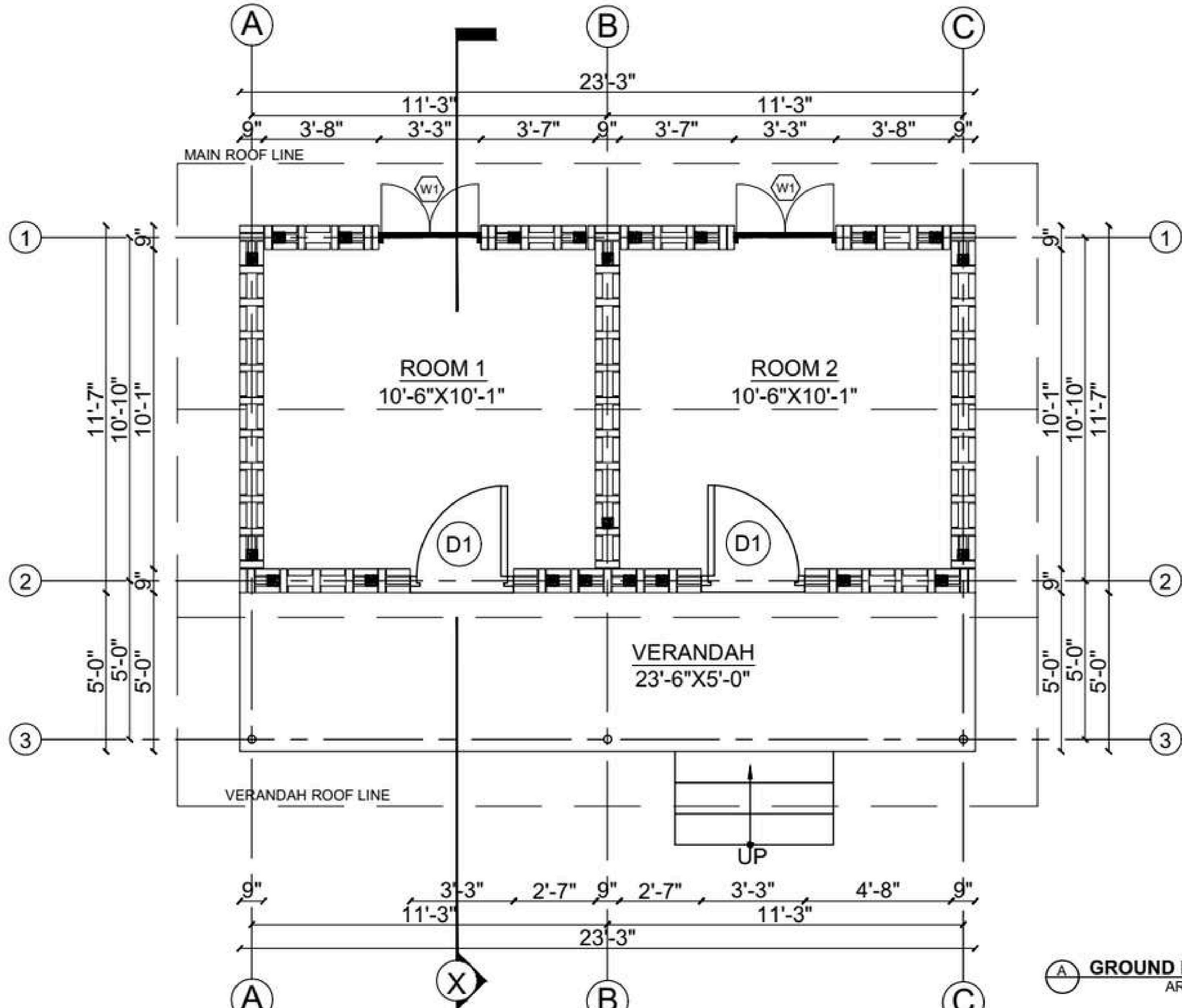
RAT-TRAP BOND MASONRY WALL CONSTRUCTION

DESIGN CONSIDERATIONS

Available Building Materials: Brick, Cement, Sand, Aggregate, Rebar, CGI Sheet
Foundation: Rubble stone masonry with cement sand mortar
Plinth: Concrete
Fence/Wall: Brick (Rat-Trap Bond) Masonry Wall
Openings: Timber frame, Plywood Shutter
Ceiling: CGI Sheet
Joints: Cement Sand Mortar
Treatment (bamboo & wood): N/A
Roof Type: Pitch Roof
Roof cover: CGI Sheet
Roof structure: Pipe truss with J-Hook for Purlins
Bracing: Vertical re-bars in walls, plinth band, sill band and lintel/roof band

Note: Rate analysis can be based on field experience

LEVEL	MANPOWER		MATERIALS								
	SKILLED	UNSKILLED	STONE	BRICK	CEMENT	SAND	AGGREGATE	CGI SHEET	METAL TRUSS	REINFORCEMENT 10mm	REINFORCEMENT 8mm
	Md	Md	cu.ft	nos.	cu.ft	cu.ft	cu.ft	bundle	kg	kg	
Up to Plinth Level	12	20	20.2	2375	62	11	2.75	-	-	190	80
Superstructure	20	30	-	5250	35	4.2	2.75	-	-	220	100
Roofing	6	6	-	-	-	-	-	4.5	540		
TOTAL	**38**	**56**	**20.2**	**7625**	**97**	**15.2**	**5.5**	**4.5**	**540**	**410**	**180**



A **GROUND FLOOR PLAN**

AREA: 386.16 SQ.FT

DESIGNED BY:

Habitat for Humanity® Nepal

PROJECT TITLE:
COMMUNITY HOUSING PROJECT

DESIGN TYPE:
RAT TRAP BRICK MASONRY IN CEMENT MORTAR

DRAWING TITLE:
FLOOR PLAN

LOCATION:
KANCHANPUR NEPAL

N S E W

DRAWN BY:
HFH NEPAL

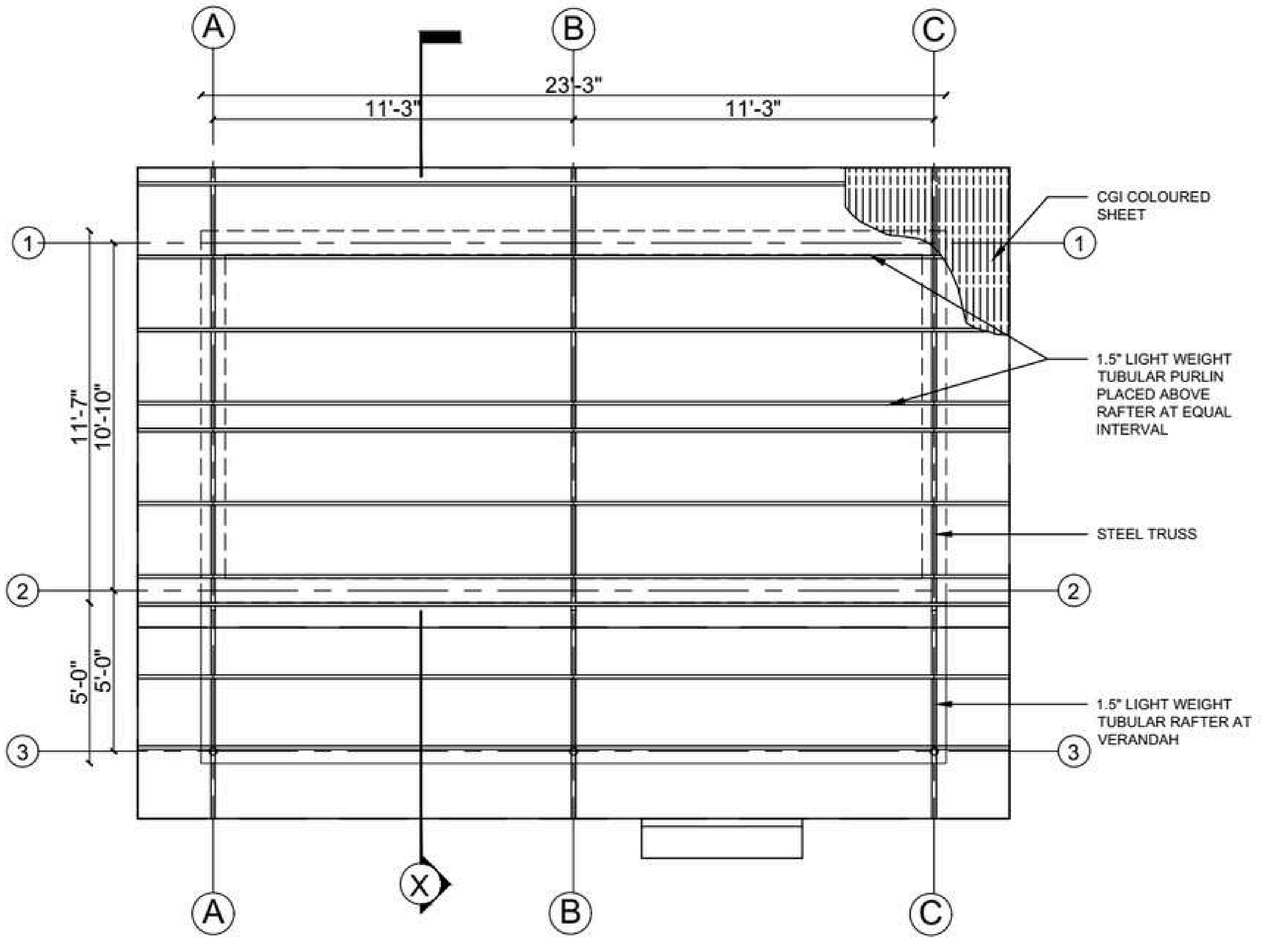
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7'-4"	2
2.	WINDOW	W1	4'-0" X 4'-3"	2

SCALE:
1:50

DATE:
2020 AD

SHEET NO:
A-01



A ROOF FRAMING DETAIL
SCALE: 1:50

DESIGNED BY:
Habitat for Humanity® Nepal

PROJECT TITLE:
COMMUNITY HOUSING PROJECT

DESIGN TYPE:
RAT TRAP BRICK MASONRY IN CEMENT MORTAR

DRAWING TITLE:
ROOF PLAN

LOCATION:
KANCHANPUR NEPAL

DRAWN BY:
HFH NEPAL

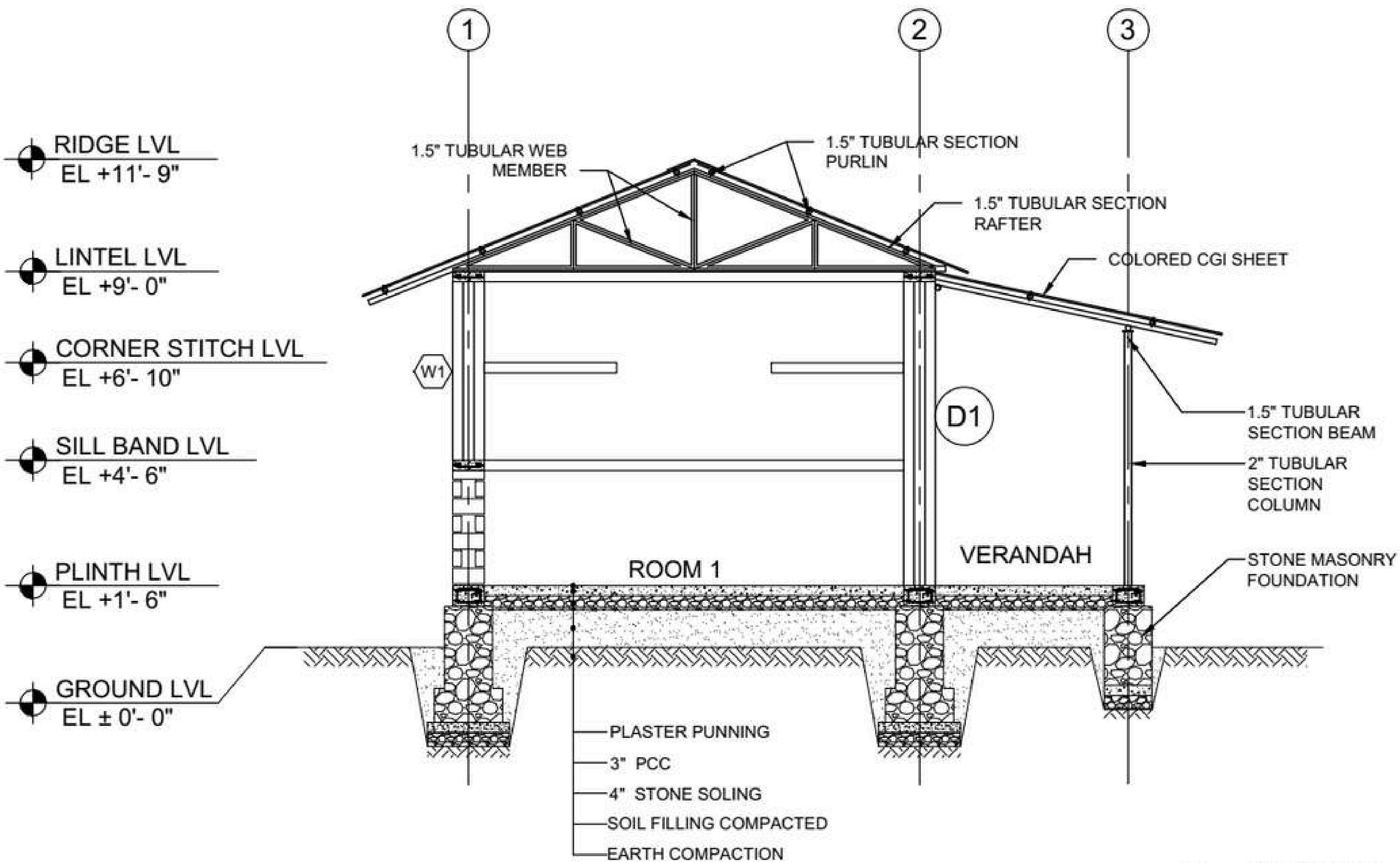
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7'-4"	2
2.	WINDOW	W1	4'-0" X 4'-3"	2

SCALE: 1:50

DATE: 2020 AD

SHEET NO: **A-02**



A

SECTION X-X
SCALE: 1:40

DESIGNED BY:
Habitat for Humanity Nepal

PROJECT TITLE:
COMMUNITY HOUSING PROJECT

DESIGN TYPE:
RAT TRAP BRICK MASONRY IN CEMENT MORTAR

DRAWING TITLE:
SECTIONAL DETAIL

LOCATION:
KANCHANPUR NEPAL

DRAWN BY:
HFH NEPAL

OPENING SCHEDULE				
S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7'-4"	2
2.	WINDOW	W1	4'-0" X 4'-3"	2

SCALE: 1:40

DATE: 2020 AD

SHEET NO: **A-03**

FRONT ELEVATION

SIDE ELEVATION

BACK ELEVATION

SIDE ELEVATION

DESIGNED BY:

PROJECT TITLE:
COMMUNITY HOUSING PROJECT

DESIGN TYPE:
RAT TRAP BRICK MASONRY IN CEMENT MORTAR

DRAWING TITLE:
ELEVATIONS

LOCATION:
KANCHANPUR NEPAL

DRAWN BY:
HFH NEPAL

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7'-4"	2
2.	WINDOW	W1	4'-0" X 4'-3"	2

SCALE: NOT TO SCALE

SHEET NO: **A-04**

DATE: 2020 AD

FLOOD RESILIENT HOUSE TYPE DESIGN VIII:

CEMENT BAMBOO FRAME TECHNOLOGY (CBFT)

INTRODUCTION

House Type: Rat-trap Bond Masonry Wall Construction
Designed By: Habitat for Humanity
Project Title: Community Housing Project
Project Location:

1. Province: Koshi
2. District: Morang
3. Ward: 12
4. VDC/ Municipality: Biratnagar Metropolitan

LOCATION MAP

CLIMATIC FEATURES:

Avg. Maximum Annual Temperature: 30.48˚C
Avg. Minimum Annual temperature: 19.53˚C
Annual Rainfall: Varies, Data from 2016-2022 shows, lowest- 1038.2mm in 2017 and highest- 2580.1 in 2020
Average relative Humidity: At 8:45am: 83.93%, At 5:45pm: 70.93%
(Data from nearby weather station)

GEOTECHNICAL FEATURES:

Topography: Plain Land of Terai
Mean Sea Level (MSL): 72 meters approx.
Soil Characteristics: Fertile Alluvial Soil
Disaster: Seasonal Flooding

WHY IS THIS DESIGN FLOOD RESILIENT?

- Plinth raised above recorded high flood level.
- Site and road level of settlement is raised with sufficient soil compaction.
- It is the settlement design including raised emergency assembly area with wheelchair accessibility, toilet, bathroom and drinking water facility.
- Building materials are used wisely: Materials not affected by water are used in substructure and bamboo is used above plinth level providing cement sand plaster and large roof overhangs for protection from rain.

The house is performing good during recent flood and water level during flood is low as site is elevated.

Habitat for Humanity® Nepal

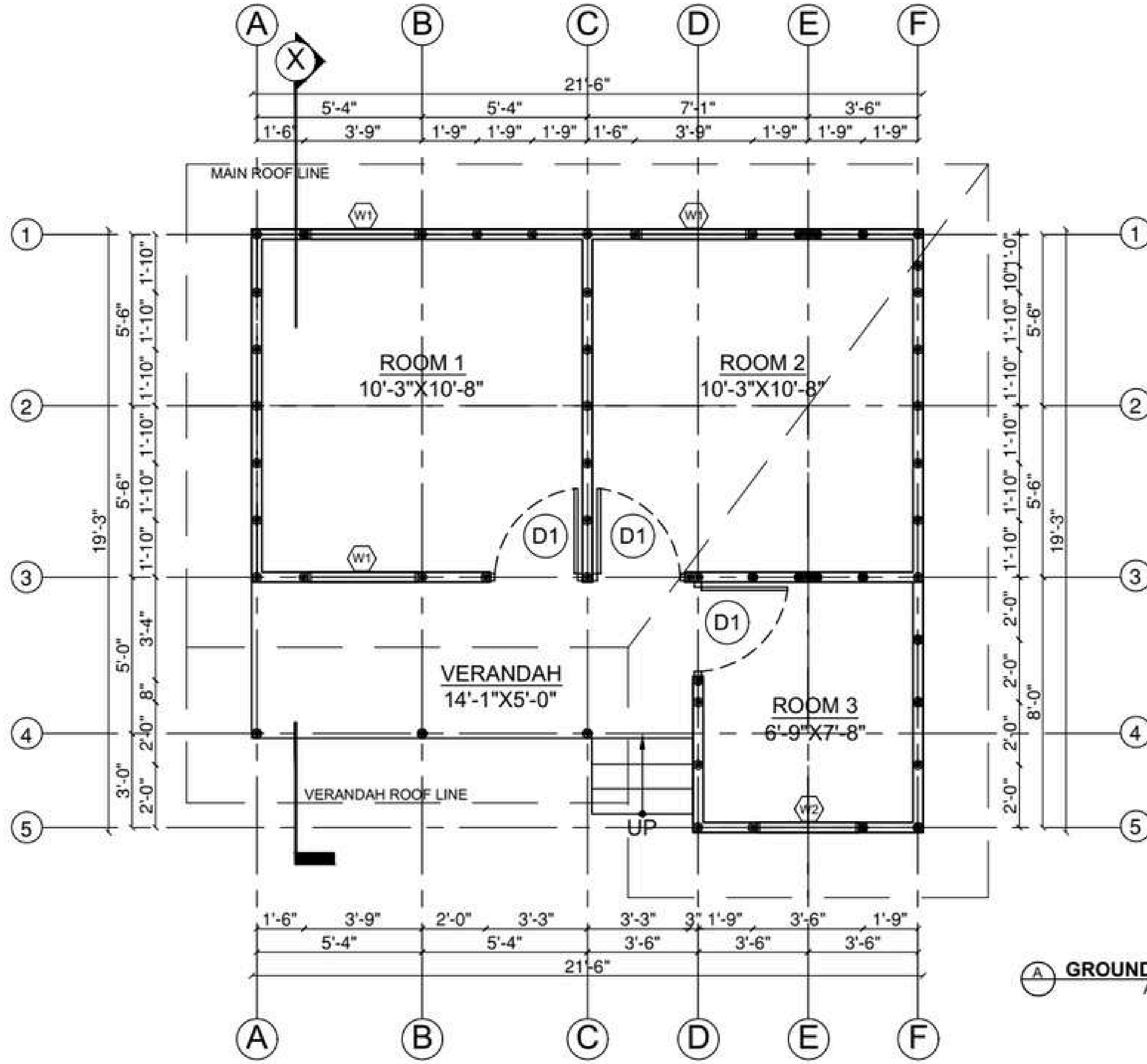
CEMENT BAMBOO FRAME TECHNOLOGY (CBFT)

DESIGN CONSIDERATIONS

Available Building Materials: Brick, Cement, Sand, Aggregate, Rebar, CGI Sheet, Bamboo
Foundation: Brick Masonry wall with pedestal column
Plinth: Concrete
Fence/Wall: Bamboo matrix with flattened bamboo cladding, a layer of chicken wiremesh and cement sand plaster
Openings: Timber frame, Plywood Shutter
Ceiling: CGI Sheet
Joints: Cement sand plaster, Nuts and bolts
Treatment (bamboo): Runnning water wash and soaking in Borax-Boric Acid Solution
Roof Type: Pitch Roof
Roof cover: CGI Sheet
Roof structure: Bamboo truss, bamboo rafters and purlins
Bracing: Diagonal bamboo bracing in roof, bambooo struts, corner strengthening of wall by metal strap, horizontal bands in foundation with pedestal column

Note: Rate analysis can be based on field experience

LEVEL	MANPOWER		MATERIALS									
	SKILLED	UNSKILLED	STONE	BRICK	CEMENT	SAND	AGGREGATE	CGI SHEET	Bamboo	Chicken mesh wire	REINFORCEM 10mm	REINFORCEM 7mm
	Md	Md	cu.ft	nos.	cu.ft	cu.ft	cu.ft	bundle	20 ft length	kg	kg	kg
Up to Plinth Level	12	6	-	1650	25	3	1.5	-	-	-	321	63
Superstructure	23	65	-	-	50	7.5	1.5	-		80	-	-
Roofing	12	12	-	-	-	-	-	4.5	200	-	-	-
TOTAL	**47**	**83**	**0**	**1650**	**75**	**10.5**	**3**	**4.5**	**200**	**80**	**321**	**63**



A **GROUND FLOOR PLAN**
AREA: 301.23 SQ.FT

DESIGNED BY:

Habitat for Humanity Nepal

PROJECT TITLE:
SAMPANNA BASTI MAHANAGR AWAS PARIYOJANA

DESIGN TYPE:
CEMENT BAMBOO FRAME TECHNOLOGY

DRAWING TITLE:
FLOOR PLAN

LOCATION:
BIRATNAGAR NEPAL

DRAWN BY:
HFH NEPAL

N E S

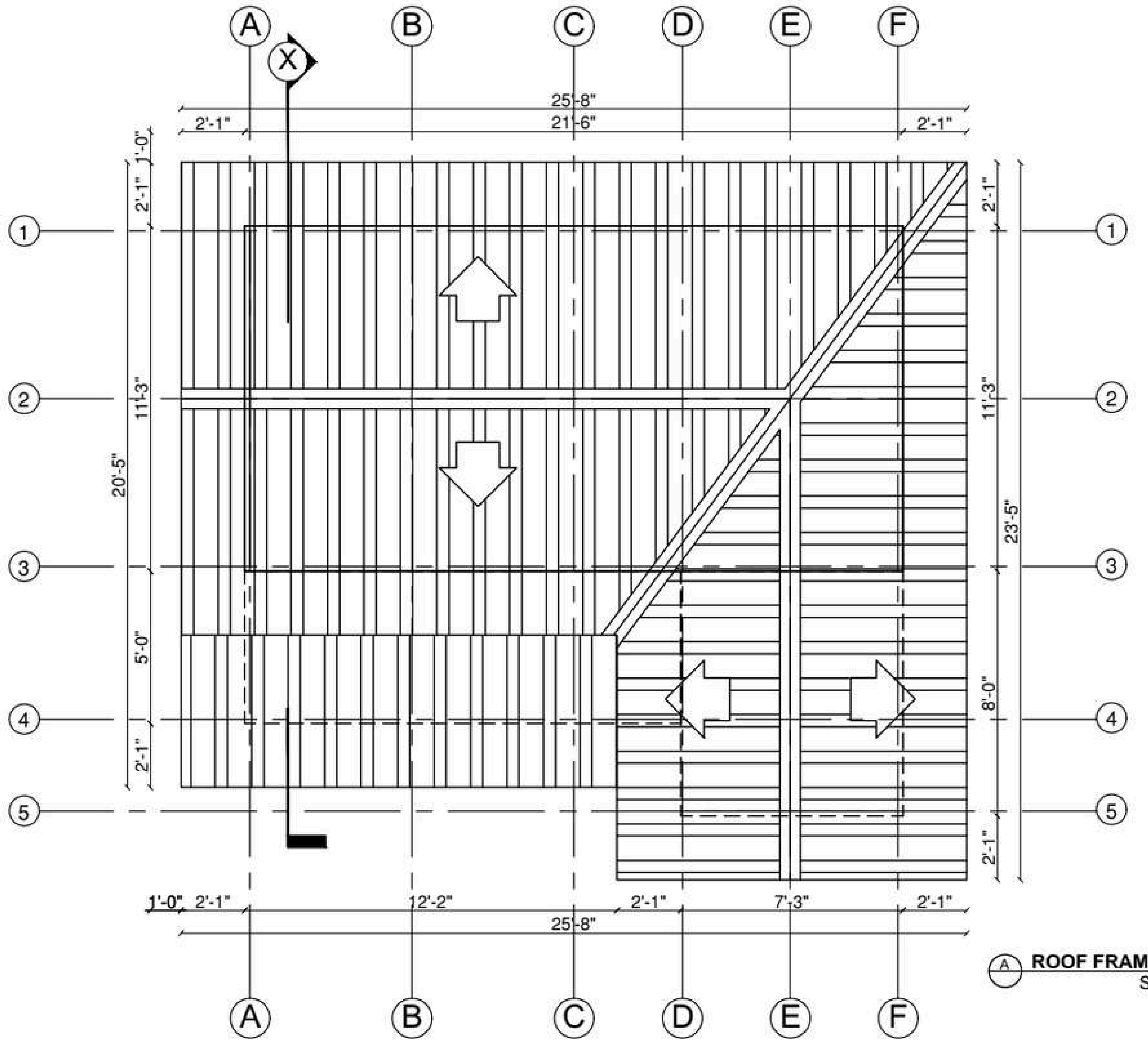
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7'-0"	2
2.	WINDOW	W1	3'-9" X 4'-0"	2
2.	WINDOW	W1	3'-6" X 4'-0"	2

SCALE:
1:50

DATE:
2021 AD

SHEET NO:
A-01



A ROOF FRAMING DETAIL

SCALE: 1:50

DESIGNED BY:

Habitat for Humanity Nepal

PROJECT TITLE:
SAMPANNA BASTI MAHANAGR AWAS PARIYOJANA

DESIGN TYPE:
CEMENT BAMBOO FRAME TECHNOLOGY

DRAWING TITLE:
ROOF PLAN

LOCATION:
BIRATNAGAR NEPAL

DRAWN BY:
HFH NEPAL

N E S W

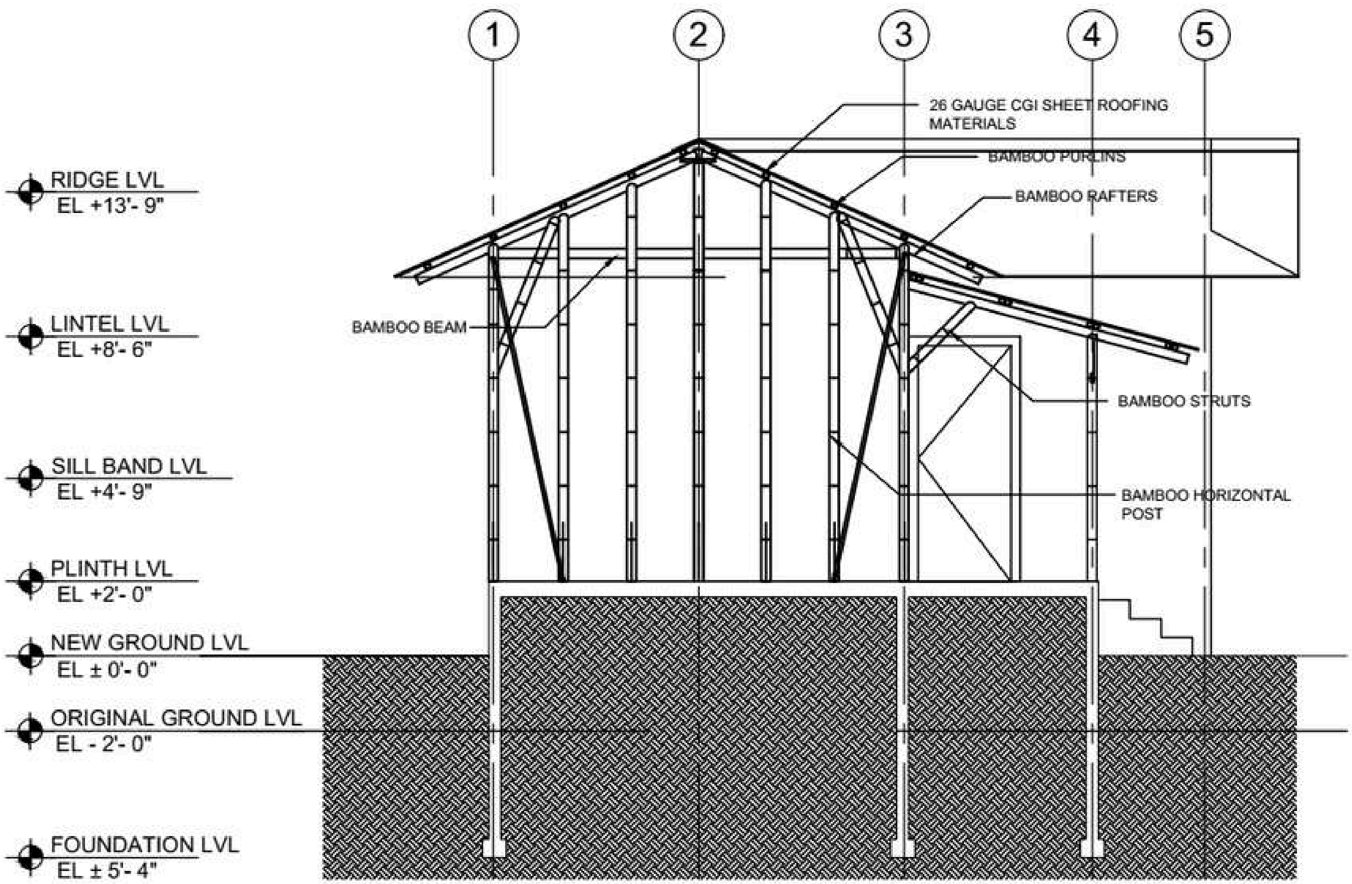
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7'-0"	2
2.	WINDOW	W1	3'-9" X 4'-0"	2
2.	WINDOW	W1	3'-6" X 4'-0"	2

SCALE: NOT TO SCALE

SHEET NO: **A-02**

DATE: 2021 AD



A

SECTION X-X
SCALE: 1:50

DESIGNED BY:
Habitat for Humanity Nepal

PROJECT TITLE:
SAMPANNA BASTI MAHANAGR AWAS PARIYOJANA

DESIGN TYPE:
CEMENT BAMBOO FRAME TECHNOLOGY

DRAWING TITLE:
SECTIONAL DETAILS

LOCATION:
BIRATNAGAR NEPAL

N
S

DRAWN BY:
HFH NEPAL

OPENING SCHEDULE

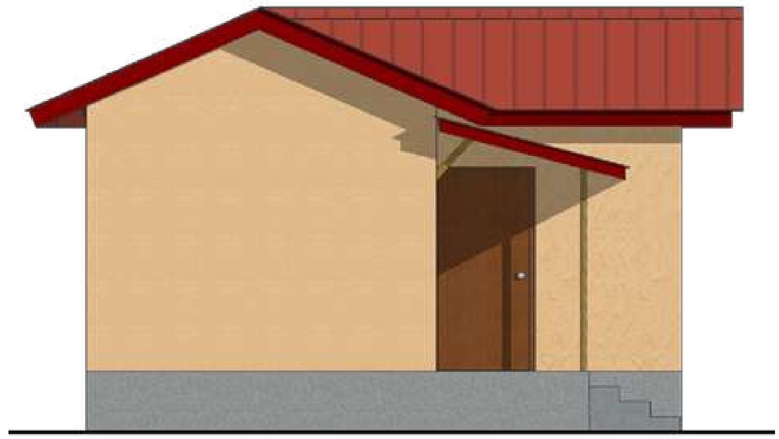
S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7-0"	2
2.	WINDOW	W1	3'-9" X 4'-0"	2
2.	WINDOW	W1	3'-6" X 4'-0"	2

SCALE:
1:50

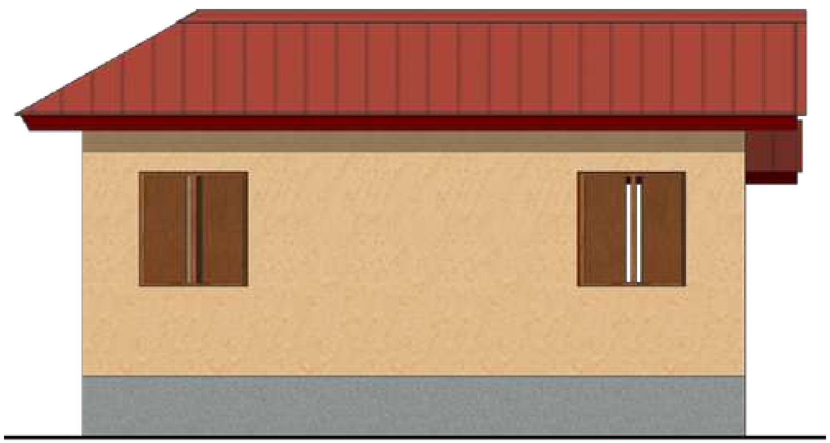
DATE:
2021 AD

SHEET NO:
A-03

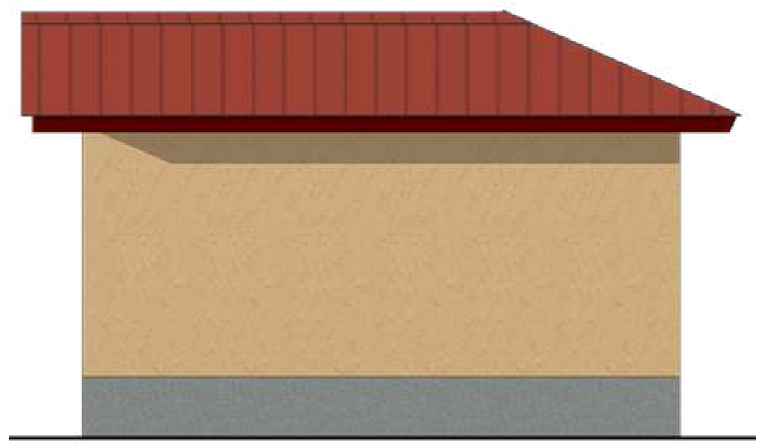
FRONT ELEVATION



SIDE ELEVATION



BACK ELEVATION



SIDE ELEVATION

DESIGNED BY:

Habitat for Humanity Nepal

PROJECT TITLE:
SAMPANNA BASTI MAHANAGR AWAS PARIYOJANA

DESIGN TYPE:
CEMENT BAMBOO FRAME TECHNOLOGY

DRAWING TITLE:
ELEVATIONS

LOCATION:
BIRATNAGAR NEPAL

DRAWN BY:
HFH NEPAL

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 7-0"	2
2.	WINDOW	W1	3'-9" X 4'-0"	2
2.	WINDOW	W1	3'-6" X 4'-0"	2

SCALE: NOT TO SCALE

DATE: 2021 AD

SHEET NO: **A-04**

FLOOD RESILIENT HOUSE TYPE DESIGN IX::

HOUSE RAISED ON STONE MASONRY WITH CSEB WALL

INTRODUCTION

House Type: Stone Masonry with CSEB Wall
Designed By: Habitat for Humanity Nepal
Project Title: Surakshit Awas Pariyojana
Project Location:

1. Province: Sudurpaschim
2. District: Kailali
3. Ward: 3
4. VDC/ Municipality: Bhajani

LOCATION MAP

CLIMATIC FEATURES:

Avg. Annual Maximum Temperature: 31.08˚C
Avg. Annual Minimum temperature: 17.1˚C
Annual Rainfall: Varies, Data from 2016-2022 shows, lowest- 824.6mm in 2019 and highest-1756.1 in 2022
Average relative Humidity: At 8:45am: 84.42%, At 5:45pm: 73.48%
(*Data from nearby weather station)*

GEOTECHNICAL FEATURES:

Topography: Plain land of terai
Mean Sea Level (MASL): 132 meters approx.
Soil Characteristics: Fertile Alluvial soil
Disaster: Seasonal inundation

WHY IS THIS DESIGN FLOOD RESILIENT?

- High plinth level, 1' above recorded high flood level.
- Toilet provided on same plinth level as that of house.
- Completely sealed septic tank to prevent water contamination and disease outbreak after the flood.
- Water resistant building materials are used.
- Sufficiently large roof overhangs are provided to protect the walls during rain.

In 2024 inundation, water level was below plinth level and people were residing inside the house.

Habitat for Humanity® Nepal

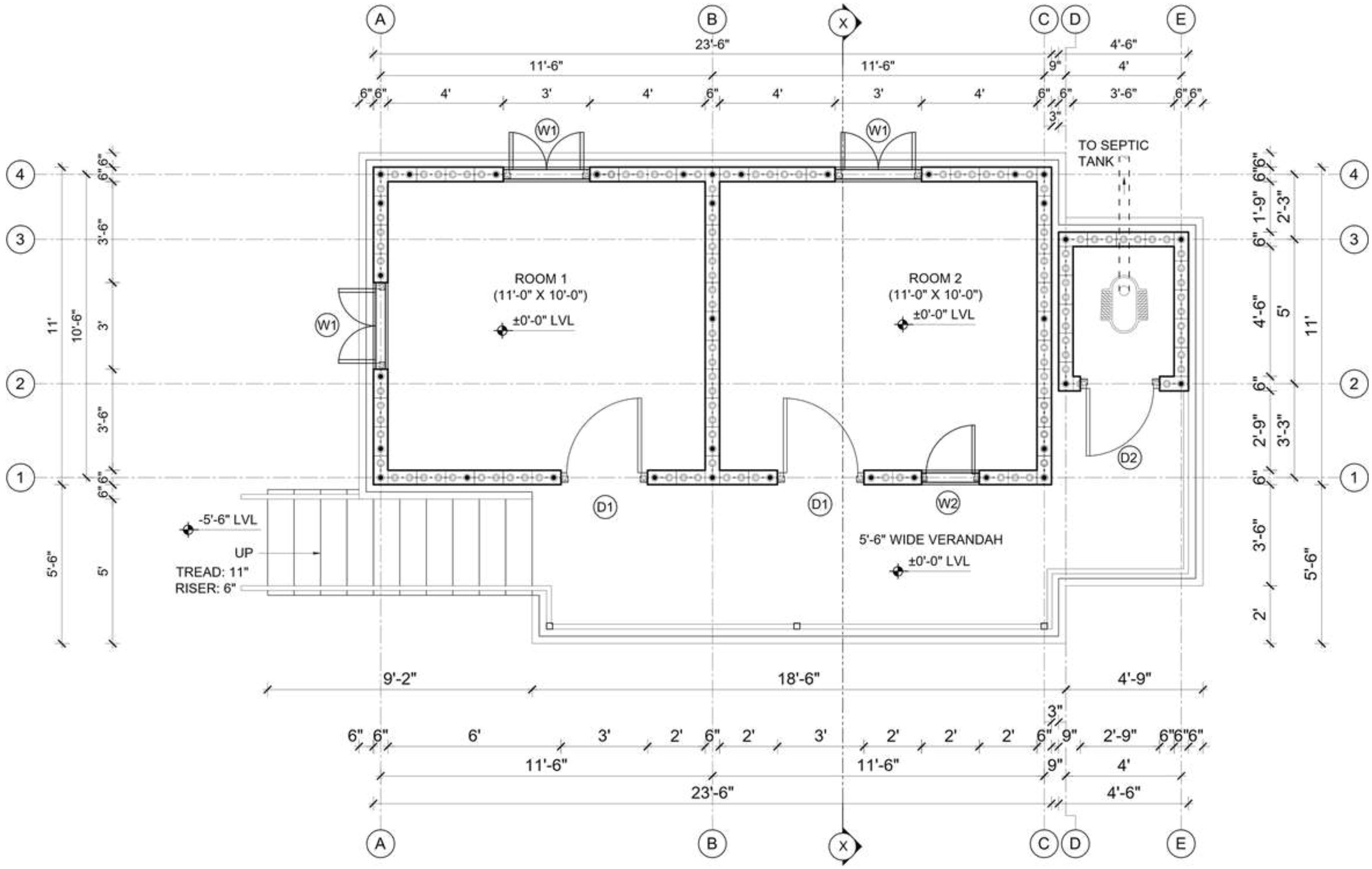
HOUSE RAISED ON STONE MASONRY WITH CSEB WALL

DESIGN CONSIDERATIONS

Available Building Materials: Cement, Sand, Soil, Aggregate, Rebar, CGI Sheet
Foundation: Stone Masonry with Cement Mortar
Plinth: Concrete
Fence/Wall: CSEB Masonry Wall
Openings: Timber frame, Plywood Shutter
Ceiling: CGI Sheet
Joints: Cement sand mortar
Treatment (bamboo & wood): N/A
Roof Type: Pitch Roof
Roof cover: CGI Sheet
Roof structure: Steel pipes rafters and purlins
Bracing: Horizontal bands at critical levels and vertical re-bars in wall

Note: Rate analysis based on field experience

LEVEL	MANPOWER		MATERIALS														
	Unskilled	Skilled	Stone	Cement	Sand	Aggregate	Reinforcement	Binding Wire	CSEB	Timber frame	Openings shutter	CGI sheet (8 ft. long)	CGI sheet (7 ft. long)	Plain sheet	J- Hook	Square MS pipe	Concrete Ring
	Md	Md	Cu. m.	Bags	Cu. m.	Cu. m.	Kg	Kg	Nos.	Cu. m.	sq. m.	Nos.	Nos.	sq. m.	Nos.	Kg	Nos.
Up to Plinth Level	233.19	58.31	47.74	266.58	16.88	2.13	-	-	-	-	-	-	-	-	-	-	7
Superstructure	66.92	19.55	-	62.61	4.32	6.04	798.37	7.17	1678.09	0.28	7.87		-	-	-	-	-
Roofing	9.40	7.93	-	-	-	-	-	-	-	-	-	14.50	22	6.52	352.89	366.34	-
TOTAL	**309.52**	**85.78**	**47.74**	**329.19**	**21.20**	**8.18**	**798.37**	**7.17**	**1678.09**	**0.28**	**7.87**	**14.50**	**22.00**	**6.52**	**352.89**	**366.34**	**7.00**



GROUND FLOOR PLAN
GROUND FLOOR AREA: 258.5 sq. ft.

DESIGNED BY:
Habitat for Humanity Nepal

PROJECT TITLE:
SURAKSHIT AAWAS PARIYOJANA

DESIGN TYPE:
CSEB HOUSE WITH STONE FOUNDATION AND CGI SHEET ROOFING

DRAWING TITLE:
FLOOR PLAN

LOCATION:
KAILALI, NEPAL

DRAWN BY:
HFH NEPAL

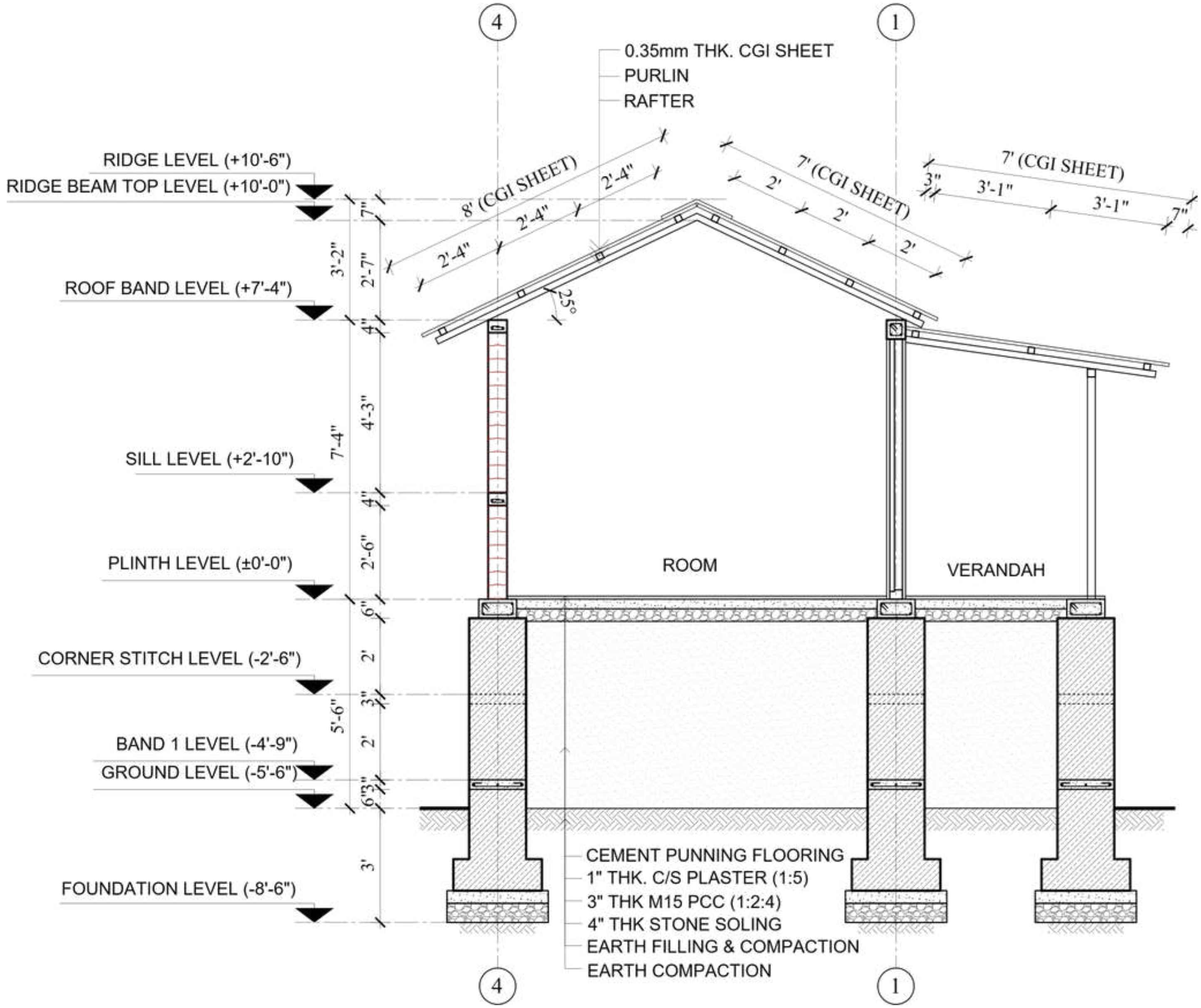
OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 6'-10"	2
2.	DOOR	D2	2'-9" X 6'-6"	1
3.	WINDOW	W1	3'-0" X 4'-3"	3
4.	WINDOW	W2	2'-0" X 4'-3"	1

SCALE: NOT TO SCALE

DATE: 2024 AD

SHEET NO: **A-01**



SECTION AT X-X

DESIGNED BY:

Habitat for Humanity® Nepal

PROJECT TITLE:
SURAKSHIT AAWAS PARIYOJANA

DESIGN TYPE:
CSEB HOUSE WITH STONE FOUNDATION AND CGI SHEET ROOFING

DRAWING TITLE:
SECTIONAL DETAILS

LOCATION:
KAILALI, NEPAL

DRAWN BY:
HFH NEPAL

N E S W

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 6'-10"	2
2.	DOOR	D2	2'-9" X 6'-6"	1
3.	WINDOW	W1	3'-0" X 4'-3"	3
4.	WINDOW	W2	2'-0" X 4'-3"	1

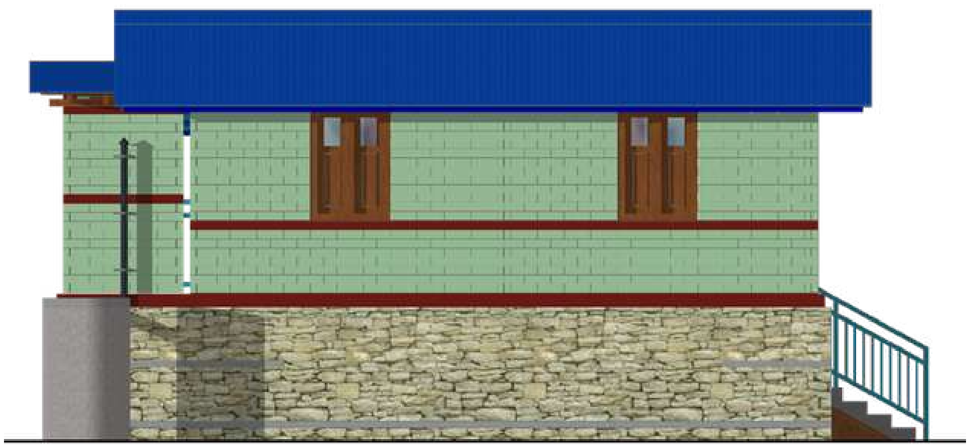
SCALE:
NOT TO SCALE

DATE:
2024 AD

SHEET NO:
A-02

FRONT ELEVATION

SIDE ELEVATION



BACK ELEVATION

SIDE ELEVATION

DESIGNED BY:

Habitat for Humanity® Nepal

PROJECT TITLE:
SURAKSHIT AAWAS PARIYOJANA

DESIGN TYPE:
CSEB HOUSE WITH STONE FOUNDATION AND CGI SHEET ROOFING

DRAWING TITLE:
ELEVATIONS

LOCATION:
KAILALI, NEPAL

DRAWN BY:
HFH NEPAL

OPENING SCHEDULE

S.N	NAME	CODE	SIZE	NOs.
1.	DOOR	D1	3'-0" X 6'-10"	2
2.	DOOR	D2	2'-9" X 6'-6"	1
3.	WINDOW	W1	3'-0" X 4'-3"	3
4.	WINDOW	W2	2'-0" X 4'-3"	1

SCALE:
NOT TO SCALE

DATE:
2024 AD

SHEET NO:
A-03

Building Safe and Dignified Homes and Communities

DEVELOPED BY

National Housing and Settlements Resilience Platform

FLOOD

Resilient Housing Solutions Technical Working Group

SCAN ME